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TOPICAL REVIEW

Advancements in Deep Learning for Automated Diagnosis of Ophthalmic Diseases: A Comprehensive Review

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ABSTRACT This review paper presents a thorough analysis of 99 recent studies focused on applying deep learning techniques for the automated diagnosis of various eye diseases, including glaucoma, diabetic retinopathy, cataracts, amblyopia, and macular degeneration. The advent of deep learning methodologies has revolutionized the field of ophthalmic diagnostics, offering promising solutions for enhancing the accuracy, efficiency, and accessibility of disease detection. The comprehensive examination encompasses diverse deep learning architectures, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid models, along with the exploration of diverse imaging modalities, such as fundus photography, optical coherence tomography (OCT), and visual field testing. Each disease category was scrutinized individually, highlighting the unique challenges and opportunities for automated diagnosis. Key findings and advancements in the field are discussed, shedding light on the potential of deep learning algorithms for early disease detection and timely intervention. This review also addresses existing limitations, including data variability, interpretability, and the need for large, diverse datasets. Insights from this literature synthesis aim to guide future research directions, fostering the continued development of reliable and efficient deep learning-based diagnostic tools for eye diseases.

INDEX TERMS Deep learning, ophthalmic diseases, glaucoma, diabetic retinopathy, cataract, amblyopia, macular degeneration, automated diagnosis, convolutional neural networks (CNNs), optical coherence tomography (OCT), fundus photography, visual field testing, disease detection, Image analysis, medical imaging, retinal diseases, neural network architectures, machine learning, healthcare technology.

I. INTRODUCTION

The field of ophthalmic diagnostics has undergone remarkable transformations with the advent of deep learning techniques. Eye diseases such as glaucoma, diabetic retinopathy, cataracts, amblyopia, and macular degeneration are leading causes of visual impairment worldwide, and early detection is critical to preventing severe vision loss. However, traditional diagnostic methods for these conditions often

require time-consuming manual analysis, are prone to inter-observer variability, and are limited by access to specialized expertise, particularly in underserved regions. This creates an urgent need for automated, accurate, and scalable diagnostic tools to support clinicians in making timely decisions.

In recent years, the integration of sophisticated algorithms, particularly deep neural networks, has shown immense potential for automating the diagnosis of these ophthalmic diseases. Deep learning, especially through convolutional neural networks (CNNs), has demonstrated significant advancements in both accuracy and efficiency, offering a

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solution to the limitations of manual diagnostic methods. The motivation for this review stems from the growing body of research in this area, yet a comprehensive analysis of these advancements has been lacking. By synthesizing findings from 101 recent research papers, this review aims to bridge that gap, providing a thorough analysis of the state-of-the-art in deep learning applications for ophthalmic disease diagnosis.

This review synthesizes findings from diverse studies that encompass various deep-learning architectures and imaging modalities. The selected studies utilized data from fundus photography, optical coherence tomography (OCT), and visual field testing, demonstrating the versatility of deep learning in processing different types of medical imaging data.

As we delve into individual disease categories, we aim to identify common trends, challenges, and opportunities within the current automated ophthalmic disease diagnosis landscape. The discussion encompasses key advancements, limitations, and potential future directions, offering a holistic view of the state-of-the-art in this rapidly evolving field of research.

This review serves as a valuable resource for researchers and practitioners interested in the intersection of deep learning and ophthalmology, guiding future research endeavors and fostering the development of robust and reliable automated diagnostic tools for improved patient outcomes.

II. METHODOLOGY

Collecting and scrutinizing papers for this review involved a systematic and rigorous approach to ensure the inclusion of relevant high-quality studies. The following steps outline the methodology employed:

A. LITERATURE SEARCH

- We conducted an extensive literature search using academic databases, such as PubMed, IEEE Xplore, ScienceDirect, and Google Scholar.
- Employed a combination of keywords, including “deep learning,” “ophthalmic diseases,” “glaucoma,” “diabetic retinopathy,” “cataract,” “amblyopia,” and “macular degeneration.”

B. INCLUSION CRITERIA

- Clear inclusion criteria were established to select studies published within a specified timeframe (e.g., the last five years).
- This study focused on papers that specifically addressed the application of deep learning techniques in the automated diagnosis of glaucoma, diabetic retinopathy, cataracts, amblyopia, and macular degeneration.

C. STUDY SELECTION

- Titles and abstracts were screened to identify potentially relevant papers.

- Studies that did not align with the scope of the review or did not involve deep learning methodologies for ophthalmic disease diagnosis were excluded.

D. FULL-TEXT REVIEW

- A thorough examination of the full text of the selected papers was conducted to assess their methodological rigor and relevance.
- The methodologies, datasets used, and reported outcomes of each study were scrutinized.

E. DATA EXTRACTION

- Key information was extracted from each selected paper, including the type of deep learning architecture employed, imaging modalities used, datasets utilized, and main findings.
- The extracted data were systematically organized to facilitate comparison and analysis.

F. QUALITY ASSESSMENT

- The quality of each selected study was assessed based on criteria such as the study design, sample size, and statistical methods.
- Gave preferences for studies with robust methodologies and well-defined evaluation metrics.

G. SYNTHESIS AND ANALYSIS

- We synthesized the findings from the selected papers and identified common themes, trends, and advancements in applying deep learning for ophthalmic disease diagnosis.
- The strengths and limitations of each study were analyzed by considering factors such as generalizability and potential biases.

H. DISCUSSION AND CONCLUSIONS

- The synthesized information is integrated into a cohesive narrative, the implications of the findings are discussed, and gaps in the current literature are addressed.
- The review was concluded by outlining key takeaways and suggesting potential avenues for future research in the field.

By rigorously following this systematic process, this review aims to provide a comprehensive and insightful analysis of the current state of deep learning applications in ophthalmic disease diagnosis based on selected 99 papers.

III. GLAUCOMA

The landscape of glaucoma diagnosis has evolved significantly with the advent of deep-learning methodologies. This review examines 25 seminal studies spanning 2018 to 2024, each contributing unique perspectives and innovations in leveraging deep learning for early glaucoma detection. The imperative to enhance diagnostic precision and expedite interventions has spurred the development of diverse models,

ranging from modified VGG16 and ResNet-50 architectures to hybrid systems integrating autoencoder networks and scale-invariant feature transforms. This introduction provides an overview of the primary objective shared by these studies: the advancement of glaucoma detection through automated methods grounded in deep-learning principles.

A. METHODOLOGY AND DATASET VARIABILITY

A close look at the research methods used in these studies reveals a variety of approaches. These studies range from *U-Net* architectures for optic cup segmentation to ensemble machine learning for optic disc and cup segmentation, each contributing novel insights to the field. Different datasets such as *ACRIMA*, *PAPILA*, *ORIGA*, and *HRF* have been utilized, showcasing a wide range of topics and facilitating comprehensive studies. This section details how the models were trained, evaluated, and the datasets used, providing a solid foundation for understanding the studies explored in more depth.

1) DEVELOPMENT OF NEW DEEP LEARNING ARCHITECTURES AND MODELS

Juneja et al. introduced GC-NET, a deep learning-based glaucoma classification network designed to automate the diagnosis of glaucoma, the second-leading cause of irreversible vision loss [10]. Their methodology involved training the network on diverse datasets to enhance its generalizability.

Kanse et al. proposed a glaucoma detection system employing a Harmonic Genetic-Based Support Vector Neural Network (HG-SVNN) classifier. To overcome drawbacks in existing methods like increased run time and complex architectures during real-time implementations, they utilized a novel hybrid feature extraction method and segmentation techniques [14]. The model was trained and tested on datasets including *ACRIMA* and *HRF*.

D'Souza et al. developed the AlterNet-K model, which outperformed other transformer and DCNN models, achieving higher classification accuracy [23]. They leveraged datasets such as *PAPILA* and *ORIGA* to train their model.

Parkhi et al. created an automated system utilizing *Deeplabv3* and ensemble machine learning for improved glaucoma identification accuracy [24]. Their approach combined optic disc and cup segmentation techniques, aligning with methodologies like *U-Net* architectures.

2) COMPARATIVE STUDIES OF DEEP LEARNING ARCHITECTURES

Shadin et al. in 2022 examined three deep-learning architectures and found that the Xception model was superior in detecting glaucoma compared to others [19]. They evaluated these models using datasets like *ACRIMA* to ensure consistency.

Raja et al. focused on utilizing different *CNN* schemes to assess their impact on performance factors such as dataset

size, architecture, and the choice between transfer learning and newly defined architectures [9]. Their study highlighted the variability in methodologies and the importance of dataset selection.

Akter et al. combined deep learning and logistic regression models to automatically and accurately detect glaucoma using features from OCT images [21]. They utilized datasets with varying characteristics to train and validate their models.

3) EARLY DIAGNOSIS OF GLAUCOMA USING DEEP LEARNING

Sudhan et al. and Kashyap R et al. studied early glaucoma detection systems. Sudhan's DenseNet-201 model achieved the highest accuracy, with 98.82% during training and 96.90% during testing [4], [5]. Their models were trained on datasets like *ORIGA* and *HRF*, emphasizing the role of dataset variability.

Ajitha S et al. created a 13-layer *CNN* highly effective in detecting glaucoma, achieving high accuracy (95.61%), sensitivity (89.58%), specificity (100%), and precision (100%) [7]. They employed datasets such as *ACRIMA* to train their models.

Shoukat A et al. in 2023 studied the use of *ResNet-50* architecture to detect glaucoma in its early stages. They obtained high accuracy, sensitivity, specificity, AUC, and F1-score across different datasets [17], showcasing the importance of diverse data in model training.

Saha et al. developed a *CNN*-based system for the automatic identification of glaucoma in 2023, achieving 91.36% accuracy, indicating its potential for early diagnosis [18]. Their methodology included training on multiple datasets to enhance model robustness.

Mahum et al. used a hybrid feature extraction method and the *Random Forest (RF) algorithm* to identify early signs of glaucoma, achieving excellent results [25]. They incorporated datasets like *PAPILA* and *ORIGA* in their study.

4) USE OF TRANSFER LEARNING AND PRE-TRAINED MODELS

Sharmila et al. leveraged deep learning with a pre-trained *Inception V3* model for image classification in glaucoma diagnosis [13]. To mitigate challenges associated with extensive training time and large dataset requirements, they utilized transfer learning. Their approach demonstrates how pre-trained models on large datasets can be fine-tuned on specific glaucoma datasets like *ACRIMA*.

5) IMAGE SEGMENTATION AND FEATURE EXTRACTION TECHNIQUES

Khan et al. studied fundus image classification, achieving up to 91.22% accuracy in classifying images of the fundus [6]. They used various datasets to train their classification models, highlighting the impact of dataset variability on performance.

S. Chandrappa et al. proposed a system employing K-means and thresholding techniques to extract the optic cup

and disc from fundus images [12]. They applied image fusion to combine optic cups and discs for improved screening. The box-counting fractal dimension estimation from fractal geometry was used to classify the fused areas, utilizing datasets such as *HRF* and *ORIGA*.

6) PREDICTIVE MODELING FOR GLAUCOMA PROGRESSION OR SURGICAL NEEDS

Wang et al. developed a Deep Learning Model (DLM) in 2024 to identify eyes likely to need surgery for uncontrolled glaucoma [15]. With an *AUC* of 0.92, the DLM effectively predicted surgery within three months, indicating its potential for timely referrals. Their methodology involved training on clinical datasets with longitudinal data to predict surgical needs.

7) APPLICATION OF DEEP LEARNING TO OCT IMAGES

Noury E et al. created a 3D deep learning algorithm using *Spectral Domain Optical Coherence Tomography (SD-OCT)* scans to detect glaucoma [20]. They aimed to demonstrate its effectiveness across real-world datasets of various ethnicities, emphasizing the importance of dataset diversity.

Akter et al. (also listed in Comparative Studies) combined deep learning and logistic regression models using *OCT* images for accurate glaucoma detection [21]. They trained their models on datasets with *OCT* imaging data, integrating methodology with dataset selection.

Jammal et al. developed a deep learning algorithm to identify errors in Retinal Nerve Fiber Layer (RNFL) segmentation on *SD-OCT* B-scans with high accuracy and sensitivity [28]. Their work highlights the significance of precise segmentation in *OCT* images and utilized specific *OCT* datasets.

8) ENHANCING INTERPRETABILITY AND TRANSPARENCY OF MODELS

Hemelings et al. worked on making deep learning models more transparent to facilitate easier detection of glaucoma [22]. They demonstrated that these models were effective even without focusing on the optic nerve head. Their methodology included explainable AI techniques and datasets that support model interpretability.

9) SYSTEMATIC REVIEWS AND COLLABORATIVE EFFORTS

Ashtari-Majlan et al. in 2023 conducted a systematic literature review of AI-based glaucoma diagnosis, focusing on recent progress and challenges [16]. This comprehensive review provides insights into collaboration between AI researchers and ophthalmologists and discusses the variability in methodologies and datasets across studies.

10) CLINICAL APPLICATIONS AND DECISION-SUPPORT SYSTEMS

Lee et al. studied test-retest variability in visual field assessments using a deep learning algorithm to predict

mean deviation from optic disc photographs [27]. This approach could serve as a valuable tool for assessing glaucoma, and they used clinical datasets to validate their models.

Christopher et al. evaluated the effectiveness of deep learning in detecting glaucomatous optic neuropathy, demonstrating its potential utility in clinical decision-support systems [26]. Their methodology involved training on large clinical datasets.

Park et al. used a *RNN* to predict visual field tests, finding it more effective than traditional methods [29]. They utilized longitudinal clinical data for training and validation.

11) ADDRESSING GLOBAL CHALLENGES IN GLAUCOMA DIAGNOSIS

Janani et al. highlighted the global significance of glaucoma as a leading cause of visual impairment, emphasizing the challenges of unidentified or untreated cases [11]. Their work underscores the need for improved diagnostic methods and increased awareness, discussing how varying methodologies and datasets can impact global health outcomes.

12) NOVEL TREATMENT METHODS FOR RETINAL VASCULAR DISORDERS

Maurya et al. in 2022 studied retinal vascular disorders and devised a new treatment method by combining a fast discrete cosine transform with neural network training [3]. This approach shows promise for blood vessel segmentation and could have implications for glaucoma-related vascular issues. They trained their models on datasets focused on retinal vasculature.

Together, these studies show how quickly deep learning and artificial intelligence help identify glaucoma. Using various imaging methods, new architectures, and clear model interpretation helps with ongoing efforts to improve early diagnosis, streamline clinical workflows, and prevent glaucoma patients from losing sight. The information in Table 1 summarizes all recent research studies that have used deep learning to identify glaucoma.

B. MODEL ARCHITECTURES AND COMPARATIVE ANALYSIS

This review aims to explore the distinct model architectures employed for glaucoma detection. This section discusses the intricacies of each model's design, from modified VGG16 and ResNet-50 to deep CNN (DCNN) such as DenseNet-201. Comparative analyses against established models such as SVM, AdaBoost, and traditional CNNs offer insights into the relative efficacy of the proposed architectures, setting the stage for a nuanced evaluation of their contributions. An in-depth examination of the evaluation metrics utilized in these studies revealed the commitment to precision, sensitivity, specificity, and other crucial parameters. Benchmarking against existing methods and across diverse datasets, such as DRISHTI, RIM-ONE, and G1020, reveals how these models fare in real-world scenarios.

TABLE 1. Overview of deep learning approaches for glaucoma detection.

Reference	Dataset Used	Adapted Methodology	Remarks	Key Findings
Saha et al. [18]	ORIGA, DRISHTI-GS, RIM-ONE v3	Modified YOLO and MobileNet	Efficient CNN-based system with high accuracy and F1 score.	High efficiency in glaucoma detection using CNN-based approaches.
Shadin et al. [19]	Fundus images	InceptionV3, ResNet50, Xception	Xception outperformed others with 96% accuracy.	Xception demonstrated superior accuracy for fundus image analysis.
Ashtari-Majlan et al. [16]	Fundus, OCT, VF images	Various deep-learning architectures	Systematic literature review, providing insights for collaboration.	Highlighted key areas for collaboration in multi-modal glaucoma research.
Hemelings et al. [22]	Fundus images	Saliency maps	Improved transparency in deep learning models for glaucoma detection.	Enhanced model transparency and explainability using saliency maps.
Parkhi et al. [24]	Fundus images	Innovative automated system for efficient glaucoma detection.	Deeplabv3 performed well in automating glaucoma detection.	
Christopher et al. [26]	Optic disc images	Deep learning algorithm	Potential utility in identifying glaucomatous optic neuropathy.	Deep learning effectively identified optic neuropathy.
Lee et al. [27]	Optic disc photographs	Deep learning algorithm	Addressed test-retest variability in visual field assessment.	Tackled variability issues in visual field assessment.
Akter et al. [21]	OCT images	DL and logistic regression	Promising results in classifying glaucomatous and non-glaucomatous eyes.	Combined DL and logistic regression improved classification.
Noury et al. [20]	SD-OCT scans	3D deep learning algorithm	High sensitivity and specificity for glaucoma detection.	3D deep learning effectively analyzed SD-OCT scans for glaucoma.
Jammal et al. [28]	SDOCT B-scans	Deep learning algorithm	High sensitivity in detecting errors in RNFL segmentation.	Detected segmentation errors in RNFL using DL models.
Singh et al. [1]	ACRIMA dataset	Modified VGG16 and ResNet-50	State-of-the-art results with 94% and 93% accuracy.	VGG16 and ResNet-50 achieved high classification accuracies.
Park et al. [29]	Visual field tests	RNNs	They predict visual field tests, offering a valuable tool for personalized patient management.	RNNs effectively predicted visual field test outcomes.
Wang et al. [15]	Not specified	Deep Learning Model (DLM)	Excellent performance with an AUC of 0.92 for predicting surgery within three months.	DLM effectively predicted the need for glaucoma surgery.
Sathya et al. [2]	Not specified	DL-EAEN (Deep Learning with feature detection)	Achieved 95.6% accuracy, outperforming existing models.	DL-EAEN outperformed other models in glaucoma detection.
Maurya et al. [3]	Not specified	Fast DCT and neural network training	Promising results in blood vessel segmentation.	Fast DCT was effective for vessel segmentation in fundus images.
Sudhan et al. [4]	Not specified	DenseNet-201	High accuracy of 98.82% in training and 96.90% in testing.	DenseNet-201 achieved high performance in glaucoma detection.
Khan et al. [6]	Not specified	CNN for fundus image classification	Maximum classification accuracy of approximately 91.22%.	CNN demonstrated high classification accuracy in fundus image analysis.
Ajitha et al. [7]	Not specified	13-layer CNN	High accuracy (95.61%), sensitivity (89.58%), specificity (100%), precision (100%).	13-layer CNN performed well in glaucoma classification.
Vanjire et al. [8]	Not specified	CNNs for automated glaucoma detection	Explored the detection of different stages of glaucoma.	CNNs were used to detect various stages of glaucoma effectively.
Shoukat et al. [17]	Not specified	ResNet-50	High accuracy, sensitivity, specificity, AUC, and F1-score for early-stage glaucoma detection.	ResNet-50 excelled in early-stage glaucoma detection.
D'Souza et al. [23]	Not specified	AlterNet-K (Transformer-based)	High classification accuracies, outperforming other models.	AlterNet-K (Transformer-based) performed better than other models.
Mahum et al. [25]	Not specified	Hybrid feature extraction	Impressive accuracy in discriminating glaucomatous and non-glaucomatous eyes.	Hybrid feature extraction significantly improved discrimination accuracy.

C. LIMITATIONS

Although the table highlights advancements in deep learning for glaucoma detection, several limitations persist. Inconsistencies in dataset specifications and the lack of standardized datasets hinder the generalizability of the models across diverse populations. Some studies lack detailed methodology descriptions, limiting their reproducibility and transparency. The absence of external validation using independent datasets raises concerns regarding model robustness. Ethical considerations, including patient privacy and consent, were not addressed comprehensively. Additionally, the challenge of

integrating deep learning models into clinical workflows and the black-box nature of certain architectures pose obstacles to their practical implementation. Standardizing validation metrics, assessing the robustness of models in diverse populations, and emphasizing early-stage detection are essential steps to overcome these limitations and enhance the clinical applicability of deep learning in glaucoma diagnosis.

D. SUMMARY

This table provides a comprehensive overview of recent deep learning approaches for glaucoma detection, showcasing

diverse methodologies and datasets. Noteworthy achievements include state-of-the-art accuracy by Singh et al. using modified VGG16 and ResNet-50 and Sathya et al., outperforming existing models with DL-EAEN. Sudhan et al. attained high accuracy using DenseNet-201, whereas Wang et al. demonstrated excellent performance in predicting surgery within three months. However, the summary reveals common challenges such as dataset heterogeneity, limited methodology details, and the need for external validation. Ethical considerations, interpretability issues, and integration of models into clinical workflows are crucial factors for successful implementation. Addressing these challenges will enhance the reliability and practicality of deep learning models for glaucoma diagnosis.

IV. DIABETIC RETINOPATHY

Diabetic Retinopathy (DR) is a serious complication of diabetes that affects the eyes. It is a progressive and potentially sight-threatening condition caused by damage to the retinal blood vessels, which are light-sensitive tissues at the back of the eye. DR is the leading cause of vision loss in working-age adults.

DR development of diabetic retinopathy is closely linked to the duration of diabetes and blood glucose control. Prolonged periods of high blood sugar can damage and weaken small blood vessels in the retina. Over time, this damage can result in various stages of diabetic retinopathy, including.

- **Non-proliferative Diabetic Retinopathy (NPDR):** In the early stages, the blood vessels in the retina may weaken and leak fluid or blood, leading to swelling and the formation of deposits. This stage may not cause noticeable symptoms; however, regular eye examination is crucial for early detection.
- **Proliferative Diabetic Retinopathy (PDR):** As the condition progresses, new abnormal blood vessels grow on the retinal surface. These vessels are fragile and prone to bleeding, which can lead to severe vision problems, including complete vision loss.
- **Diabetic Macular Edema (DME):** Swelling of the macula, the central part of the retina responsible for sharp central vision, can result from fluid leakage. DME is a common complication of DR and can lead to a significant loss of vision.

Regular eye examinations, including specialized tests such as fundus photography OCT, are crucial for early detection and management of diabetic retinopathy. Early intervention can help prevent or slow the progression of the disease and reduce the risk of vision loss.

A. METHODOLOGY AND DATASET VARIABILITY

Diabetic Retinopathy (DR) is a serious eye condition resulting from diabetes, potentially leading to significant vision impairment. Understanding the causes, stages, risk factors, and effects of DR is crucial for patients with

diabetes to prevent severe eyesight issues. Early detection and intervention are essential in managing DR effectively.

Recent years have witnessed numerous studies that have enhanced our knowledge of DR, exploring molecular mechanisms, innovative diagnostic methods, and improved treatments. By synthesizing these research findings, we aim to provide a comprehensive understanding of DR, aiding healthcare professionals and individuals with diabetes in addressing this eye condition. This section delves into the complexities of DR, supported by the latest research, to empower individuals in protecting their eye health.

1) DEEP LEARNING ARCHITECTURES FOR DR DETECTION

Several studies have focused on leveraging deep learning (DL) architectures for accurate detection and classification of DR stages:

- **S. Mishra et al.** utilized the Kaggle APTOS dataset with the DenseNet model, achieving an impressive accuracy of 0.9611 in classifying DR stages [30].
- **Das et al.** evaluated 26 cutting-edge DL architectures for automatic DR detection, highlighting the importance of model selection. They found EfficientNetB4 to be the most accurate [34].
- **Mohanty et al.** proposed hybrid DL architectures for early DR detection, with DenseNet121 yielding superior results, demonstrating the efficacy of DL in accurate identification and classification of DR [35].
- **Kalyani et al.** introduced a novel capsule network capable of automatically detecting and classifying different stages of DR, showing promising results in accurate identification [32].
- **Bhandari et al.** explored soft computing approaches, including Particle Swarm Optimization (PSO) and genetic algorithms, for automated DR grading, suggesting that DL methods can enhance DR detection systems [33].
- **Doshi et al.** implemented GPU-accelerated DCNN for automatic diagnosis and staging of DR. Combining three similar models improved performance, indicating the potential of DL in disease staging [39].
- **Hemanth et al.** proposed a mixed method combining image processing and DL for DR detection, achieving high accuracy and sensitivity in retinal fundus images [40].
- **Usman et al.** designed a model utilizing DL to extract and classify multi-label features for accurate DR detection, demonstrating applicability in clinical settings and large-scale screening programs [42].

2) INNOVATIVE METHODS AND SEGMENTATION TECHNIQUES

Advancements have also been made in developing new segmentation and classification methods for DR:

- **Sivapriya et al.** introduced a novel DR segmentation and classification method, effectively separating vessels

and classifying them into multiple groups. The proposed method outperformed existing techniques, indicating its potential for early DR identification and grading [36].

- **Atlas et al.** suggested using DL to identify retinal hemorrhages early, a critical aspect of DR. Their framework demonstrated high sensitivity and specificity on the DIARETDB2 dataset, supporting its use in accurate pattern recognition and classification [37].
- **Islam et al.** developed a DL method for detecting microaneurysms, achieving state-of-the-art performance with high sensitivity and specificity in early-stage DR detection [45].
- **Chetoui et al.** utilized conceptual models and different texture characteristics alongside Support Vector Machines (SVM) for classification. Their LESH method showed high accuracy and AUC in DR stage sorting [44].
- **Özbay** employed the Artificial Bee Colony (ABC) algorithm for image segmentation and introduced an active DL approach. The ADL-CNN model demonstrated excellent performance in identifying DR lesions [46].

3) TRANSFER LEARNING AND HYBRID MODELS IN DR DETECTION

The application of transfer learning and hybrid models has shown significant promise:

- **Shoaib et al.** addressed the urgent need for early DR detection by employing advanced DL techniques and transfer learning, resulting in improved accuracy and potential to transform DR diagnosis [49].
- **Bosale et al.** combined transfer learning with CNNs to enhance DR detection. Their method performed well in detecting severe visual impairment, achieving a high quadratic weighted kappa score in the APTOS 2019 Blindness Detection Competition [50].
- **Al-Ahmadi et al.** explored DR diagnosis in the Middle East using DL and transfer learning algorithms. The DenseNet121 model proved highly effective, showcasing the automation potential in DR diagnosis [51].
- **Al-Absi et al.** introduced DiaNet v2 to improve diabetes diagnosis accuracy in the MENA region. Utilizing DL on retinal images, DiaNet v2 distinguished between diabetic patients and controls with high accuracy [47].

4) AUTOMATED CLASSIFICATION SYSTEMS AND HYBRID APPROACHES

Several studies have proposed automated systems and hybrid approaches for DR detection:

- **Saranya et al.** proposed an automated model combining pre-processing, semantic segmentation, and severity classification for early DR identification. The method aimed to enhance diagnosis speed and accuracy across multiple datasets [31].
- **Nguyen et al.** addressed the global health issue of DR by suggesting an automated classification system

employing various machine learning models. The system effectively classified DR severity grades, highlighting the importance of rapid detection [41].

- **Qureshi et al.** utilized an active DL model, the updated ADL-CNN, achieving high sensitivity, specificity, and accuracy in DR stage recognition. The model also facilitated easier recognition of DR-related lesions [43].
- **Gangwar et al.** introduced a new DL hybrid model focusing on individuals aged 20–65 years. The model outperformed other published results, indicating its potential in addressing DR within this demographic [38].

5) IMPORTANCE OF EARLY DETECTION AND MACHINE LEARNING ALGORITHMS

The critical role of early detection and the application of machine learning algorithms have been emphasized:

- **D. A. Hasan et al.** investigated the use of machine learning algorithms to detect and classify DR by training on large public datasets of retinal fundus and thermal images. The ResNet50 deep CNN performed best in identifying warning signs and assessing DR severity through effective feature extraction [52].
- **Dai et al.** developed DeepDR Plus, a DL system with personalized screening intervals to predict DR progression. The system showed high concordance indices and Brier scores, suggesting its utility in individualized risk prediction [48].
- **Islam et al.** conducted a systematic review and meta-analysis to evaluate the effectiveness of DL algorithms in identifying and classifying DR. The study demonstrated that DL algorithms possess high precision, sensitivity, specificity, and AUC, supporting their use as automated tools in accurate DR assessment and screening programs [53].

The information from these studies adds to the knowledge about automated DR detection. They show how new technologies can change the diagnosis and treatment of dangerous diabetes complications. In Table 2, studies on the automated detection of diabetic retinopathy are summarized.

B. MODEL ARCHITECTURE AND COMPARATIVE ANALYSIS

The automated detection of diabetic retinopathy (DR) is contingent upon the robustness of the model architectures, and diverse approaches have been explored in recent studies. Mishra et al. achieved a remarkable 0.9611 accuracy with a DenseNet model trained on fundus images, while Saranya et al. employed a UNet architecture for red lesion segmentation with high specificity (99%) and sensitivity (89%). Kalyani et al. proposed a reformed capsule network, showcasing impressive accuracies for different DR stages. Comparative analyses revealed the effectiveness of soft computing approaches (Bhandari et al.), optimal performance of EfficientNetB4 (Das et al.), and significance of a modified U-Net architecture (Sivapriya et al.) for vessel segmentation.

TABLE 2. Summary of studies on automated diabetic retinopathy detection.

Reference	Dataset Used	Adapted Methodology	Remarks	Key Findings
Mohanty et al. [35]	et APTOS 2019 Blindness Detection Kaggle Dataset	Hybrid network combining VGG16 and XGBoost Classifier, DenseNet 121 model	Achieved 97.30% accuracy for early DR detection.	DenseNet 121 demonstrated superior performance for early detection.
Gangwar et al. [38]	et APTOS 2019 datasets	Transfer learning on Inception-ResNet-v2, custom CNN layers	72.33% and 82.18% test accuracy on Messidor-1 and APTOS datasets, respectively.	Custom block of CNN layers enhanced DR classification accuracy.
Bosale et al. [50]	APTOS 2019 Blindness Detection Competition dataset	Transfer learning with CNNs, U-Net for segmentation	Achieved a quadratic weighted kappa score of 0.92546.	Integrated pre-trained models to boost DR detection performance.
Mishra et al. [30]	Kaggle dataset APTOS with 3662 fundus images	DenseNet model for DR stage classification	96.11% accuracy and a quadratic weighted kappa score of 0.8981.	DenseNet accurately classified DR stages.
Das et al. [34]	Kaggle's EyePACS fundus image dataset	Evaluation of 26 DL CNN architectures	EfficientNetB4 achieved a training accuracy of 99.37%.	EfficientNetB4 was the most optimal model for DR detection.
Özbay et al. [46]	EyePACS dataset	Active Deep Learning-CNN with ABC algorithm	Achieved 99.66% classification accuracy.	ABC algorithm enhanced CNN's ability in DR classification.
Hasan et al. [52]	Kaggle's EyePACS dataset	Evaluation of ML algorithms for DR detection	ResNet50 showed superior performance in identifying severity levels of DR.	ResNet50 demonstrated advanced DR classification capabilities.
Islam et al. [53]	Kaggle's EyePACS dataset	Systematic review and meta-analysis of DR detection	Emphasized the need for standardized datasets and benchmarking metrics.	Advocated for standardized benchmarks in DR research.
Al-ahmadi et al. [?]	et Kaggle's EyePACS and Messidor-1 dataset	DL and transfer learning algorithms	DenseNet121 exhibited the highest accuracy.	DenseNet121 effectively automated DR diagnosis.
Hemanth et al. [40]	et MESSIDOR database	Image processing techniques, CNN for diagnosis	Achieved 97% accuracy and 94% sensitivity.	Efficient model for classifying retinal fundus images.
Saranya et al. [31]	et IDRID and MESSIDOR datasets	UNet for segmentation, CNN for classification	Achieved high specificity (99%) and sensitivity (89%) for red lesion detection.	Excellent performance in segmenting and classifying lesions.
Kalyani et al. [32]	Messidor dataset	Reformed capsule network for automated DR detection	Achieved high accuracies (97.98%, 97.65%, 97.65%, 98.64%) for classifying DR stages.	Capsule network effectively classified different DR stages.
Sivapriya et al. [36]	et STARE, DRIVE, and Messidor-2 datasets	Modified U-Net architecture (ResEAD2Net)	Achieved 98.07% and 97.55% segmentation accuracy on Messidor-2 dataset.	Improved multiclass classification performance with U-Net.
Atlas et al. [37]	DIARETDB2 dataset	ELSTM CNN, MSER for retinal hemorrhage detection	Achieved 98.67% sensitivity and 98.91% specificity.	Highly effective for hemorrhage detection using ELSTM CNN.
Usman et al. [42]	Color Fundus Photographs (CFPs)	DL Multi-Label Feature Extraction, ResNet50, SqueezeNet1	ResNet50 achieved 93.67% accuracy, SqueezeNet1 also performed well.	Effective for large-scale DR screening.
Al-Absi et al. [47]	et Qatar Biobank and Hamad Medical Corporation datasets	DiaNet v2 deep learning model for diabetes detection	Achieved over 92% accuracy, 93% sensitivity, and 91% specificity.	DiaNet v2 was effective in detecting diabetes in large datasets.
Shoaib et al. [49]	Ocular Disease Intelligent Recognition (ODIR)	Transfer learning with Inception-ResNetv2, customized DiaCNN	DiaCNN achieved 100% accuracy in training and 98.3% in testing.	DiaCNN demonstrated outstanding accuracy in DR detection.
Islam et al. [45]	Kaggle diabetic retinopathy dataset	Deep CNN for microaneurysm detection	Achieved state-of-the-art performance with a kappa score of 0.851 and AUC of 0.844.	CNN achieved excellent results for microaneurysm detection.
Doshi et al. [39]	Not specified	GPU-accelerated DCNNs for automated DR diagnosis	Ensembling three models yielded a score of 0.3996 on the quadratic weighted kappa metric.	Ensembling CNN models enhanced DR detection performance.
Nguyen et al. [41]	Not specified	CNN, VGG-16, VGG-19 for fundus image analysis	80% sensitivity, 82% accuracy, 82% specificity, 0.904 AUC in DR severity grading.	VGG architectures performed well in categorizing DR severity.
Qureshi et al. [43]	Not specified	Active Deep Learning-CNN for DR stage recognition	Achieved 92.20% sensitivity, 95.10% specificity, and 98% accuracy.	Active DL-CNN excelled in lesion recognition for DR.
Chetoui et al. [44]	Not specified	LESH with SVM-RBF for DR detection	Best accuracy of 90.04%, AUC of 93.01%.	LESH-SVM-RBF performed well in detecting DR.

Transfer learning models, such as Inception-ResNet-v2 (Gangwar et al.), have demonstrated superior outcomes, and the exploration of DCNNs (Hemanth et al.) and machine learning models (Nguyen et al.) have shown promising

results. The field has also witnessed advancements with active deep learning CNN (Qureshi et al.) and DiaNet v2 (Al-Absi et al.), emphasizing the potential for extending screening intervals (Dai et al.). These diverse architectures

collectively contribute to the ongoing refinement of automated DR diagnosis, highlighting the accuracy, sensitivity, and specificity of the various methodologies.

C. LIMITATIONS

Despite notable advancements in automated diabetic retinopathy (DR) detection, several limitations persist across studies. One common constraint is the lack of standardized datasets, as different research works employ varied datasets, such as Kaggle's EyePACS, APTOS, and Messidor, posing challenges in directly comparing outcomes and generalizing results. Additionally, variations in image quality, resolution, and annotation protocol contributed to inconsistencies. The limited diversity in the geographical origin of the datasets and patient demographics may affect the robustness of the models across diverse populations. Moreover, the scarcity of real-world clinical implementations and validation in large-scale heterogeneous populations impedes the translation of these models into practical healthcare settings. Addressing these limitations is crucial to enhance the reliability, generalizability, and real-world applicability of automated DR detection systems.

D. SUMMARY

In summary, the landscape of automated diabetic retinopathy (DR) detection has undergone significant progress with the adoption of diverse model architectures and methodologies. The reviewed studies showcase the potential of deep learning, transfer learning, and innovative CNN architectures in achieving high accuracy, sensitivity, and specificity in detecting different stages of DR. While each study makes unique contributions, the lack of standardized datasets and limited real-world validation pose challenges to the field's advancement. Efforts to standardize datasets, enhance model robustness across diverse populations, and validate findings in real-world clinical settings are imperative. Continuous collaboration between researchers, clinicians, and technology developers is essential to bridge existing gaps and propel automated DR detection systems for practical implementation, ultimately improving the early diagnosis and management of this critical condition.

V. CATARACT

Cataract, a prevalent age-related ocular condition, represents a significant global health concern and a leading cause of vision impairment and vision loss. Characterized by clouding of the eye's natural lens, cataracts hinder light transmission, resulting in blurred vision and diminished visual acuity. The development of cataracts is often associated with aging; however, other factors, such as prolonged exposure to ultraviolet radiation, certain medications, trauma, or systemic diseases, can contribute to their onset. As the aging population continues to grow, the incidence of cataracts is anticipated to increase, emphasizing the importance of understanding, diagnosing, and effectively managing this condition. With advancements in surgical techniques and

intraocular lens (IOL) technology, cataract treatment has become highly successful, restoring visual clarity and significantly improving the quality of life of affected individuals.

A. METHODOLOGY AND DATASET VARIABILITY

The methodology employed in studies related to cataract detection and variability in datasets plays a pivotal role in shaping the outcomes and applicability of automated diagnostic systems. Various methodologies have been utilized, ranging from traditional image-processing techniques to advanced deep-learning approaches.

Image datasets, which are crucial for training and evaluating these systems, exhibit considerable variability in size, diversity, and annotation protocol. Some studies leverage curated datasets with well-defined cataract classes, whereas others use real-world clinical images with inherent complexities.

The choice of methodology and dataset profoundly affects the robustness, generalizability, and real-world applicability of automated cataract detection models. This section explores the diverse methodologies and dataset variations encountered in the development of effective and reliable systems for cataract diagnosis.

B. MODEL ARCHITECTURE AND COMPARATIVE ANALYSIS

Many recent studies have examined how machine and deep learning algorithms can automatically sort cataract stages during surgery. Yu F et al. used a dataset of 100 cataract surgeries to compare support vector machine (SVM), RNN, and CNN algorithms. Their results showed that deep learning methods, particularly the image-only CNN-RNN model, are very good at obtaining accurate results for automated phase detection during cataract surgery [54]. Another important contribution comes from J. Ran et al., who suggested using fundus images to diagnose cataracts. By combining a DCNN and Random Forests (RF), they achieved an average accuracy of 90.69% for six-level cataract grading, which is better than that of traditional four-level grading systems. The study emphasized the importance of using a nuanced grading system to obtain a full picture of patient conditions [55]. To help identify cataracts early, Yusuf et al. created a web-based Computer-Aided Diagnostic (CAD) system that uses a CNN. The CNN performed well when trained on a set of 100 eye images, with a sensitivity of 69%, specificity of 86%, and overall accuracy of 78%. The system made it easy for people who were not doctors to find cataracts early, which is useful outside of clinical settings [56]. To address the lack of eye doctors in rural China, Zhang et al. suggested a six-level method for grading cataracts using multi-feature fusion. With ResNet18 and the texture features built into their algorithm, they achieved an impressive average accuracy of 92.66%. The study showed that it might be possible to make cataract diagnosis more accurate and faster, especially for people who do not receive sufficient care [57]. Simanjuntak et al. compared four CNN architectures to determine how cataracts

can be detected. To help doctors identify cataracts earlier, they were divided into four groups. The proposed CNN model exhibited the best and most stable performance with a 92% success rate. This study focused on adding more datasets to improve the classification accuracy [58]. Wang et al. used the Collaborative Monitoring Deep Learning (CMDL) method to look for cataracts in ocular B-ultrasound images. This method, which uses YOLO-v3 and feature extraction modules, worked well in improving the accuracy of cataract detection, especially when dealing with problems such as strong echoes and low contrast [61]. Huang et al. created DCNN models that accurately identify and grade nuclear cataracts using AS-OCT images from the anterior segment. Their end-to-end DCNN worked better than the others, giving doctors a reliable way to track the worsening of cataracts [62]. In 2023, Shetty et al. suggested a VGG16-based model for identifying cataracts, aiming to improve the problems with the current systems. At each layer, their model jumbled and combined images, making it easy for people in rural areas to determine if they had a cataract without seeing an eye doctor [61]. In their 2023 paper, Khan et al. introduced CataractNet, which automatically finds cataracts in fundus images. Using small kernels and optimizing parameters, they made their network much faster and cheaper to run and achieved an impressive average accuracy of 99.13% [64]. G. ARSLAN et al. used deep learning models to study several eye diseases, such as cataracts. Using CNNs, their study showed how well the EfficientNet architecture worked, providing a strong way to quickly find eye diseases [63]. Faizal et al. developed another creative idea: an algorithm that can automatically detect cataract disease in both visible wavelength and anterior segment images. The algorithm, which used adaptive thresholding and a fine-tuned CNN model, correctly classified approximately 95% of cases, finding either a nuclear cataract, a cortical cataract, or a hybrid case involving both types of cataracts [66]. Varma et al. used fundus image analysis to make an automatic system for classifying and grading cataracts that would work well and not cost too much. Their DCNN was 92.7% accurate in classifying and grading cataracts in four stages, which shows the accuracy of cataract diagnosis [67]. Yadav et al. discussed the problems of finding cataracts early and suggested a mixed method using CNNs that have already trained and supported vector machine classifiers. Ensemble learning worked very well, correctly classifying cataracts into four groups with 96.25% accuracy, and it appears that this could be a good way to make accurate early stage diagnoses [68]. The GDCNN algorithm was created by Manikandan et al. as a way to make a good tool for identifying cataracts. Using a Generative Adversarial Network (GAN) and DCNN, their combined method achieved 99.21% accuracy, beating the best methods available at the time [67]. In response to problems worldwide, Yadav et al. suggested a cheap and useful way to find cataracts early on. When they combined deep learning with the 2D-discrete Fourier transform (DFT) spectrum of fundus images,

their system achieved a remarkable four-class accuracy of 93.10%, showing that it was better than the best methods used previously [68]. Jones et al. wanted to find a way to prevent cataracts from causing severe visual impairment. They developed a DCNN model based on GoogleNet architecture. After training on 66 images, the model showed good results, with an overall training accuracy of 86.9% and a validation accuracy of 35.8% [69]. Pratap et al. added to the important topic of finding cataracts early by suggesting a computer-assisted automatic detection method that uses images of the fundus retina. Their method had a four-stage classification accuracy of 92.91%, demonstrating the effectiveness of the proposed approach for classifying cataracts [70]. Sahayam et al. used an EfficientNet-based CNN to try to automate the diagnosis of cataracts using fundus images. The proposed CNN method showed promise for quick and accurate diagnosis of cataracts, with high dice scores, sensitivity, and accuracy on a validation dataset [71]. Chauhan et al. looked at the worldwide problem of cataracts causing severe vision loss and suggested using computers to help with early detection. Their group model, which included the VGG-19, ResNet101V2, and InceptionV3 models, obtained an impressive F-1 Score of 95.90 on the test dataset, showing that it was good at correctly diagnosing cataracts [72]. Angeline et al. studied how to find cataracts early by taking pictures of the front of the eye with a smartphone camera app. The system detected cataracts with 92.1% accuracy using the pre-trained VGG16 model as a CNN architecture. The method aims to make traditional fundus images easier and less expensive, making early detection and treatment easier for more people [73].

Table 3 demonstrates the growing success of machine learning and deep learning approaches in automating cataract detection and classification. These advanced algorithms exhibit high accuracy and provide valuable tools for early diagnosis and intervention. The diversity of methods, from web-based systems to deep neural networks, underscores the versatility of these technologies for addressing the global challenge of cataract-related vision loss. Continued research and innovation in this field promises to improve the precision and accessibility of cataract diagnosis, ultimately enhancing patient outcomes and preventing vision loss.

C. LIMITATIONS

Despite promising advancements in machine learning and deep learning for cataract detection, several limitations persist across the reviewed studies. First, reliance on datasets of varying sizes and sources may introduce biases and affect the generalizability of the models. Limited diversity in the datasets, especially regarding demographic representation and cataract severity, may affect the robustness of the algorithms across different populations. Additionally, some studies lack detailed explanations of the interpretability and transparency of their models, posing challenges in understanding decision-making processes. The need for large, diverse datasets, transparent model architectures, and

TABLE 3. Comprehensive overview of cataract detection.

Reference	Dataset Used	Adapted Methodology	Remarks	Key Findings
Wang et al. [59]	1,309 AS-OCT images	Collaborative Monitoring Deep Learning	0.97 AUC, Precise detection and grading	High AUC score for precise detection and grading
Huang et al. [60]	1,309 AS-OCT images	DCNN, DCNN+RF	0.97 AUC, Precise detection and grading	Consistent AUC score with Wang et al. (2021)
Yadav et al. [65]	Fundus retinal images	CNN (AlexNet, VGGNet, ResNet)	96.25% accuracy, Ensemble learning	High accuracy with ensemble learning approach
Yadav et al. [68]	Fundus images	DL with 2D-DFT spectrum	93.10% accuracy, Affordable and efficient system	Affordable and efficient cataract detection system
Simanjuntak et al. [58]	399 fundus images	GoogleNet, MobileNet, ResNet, Proposed CNN	92% to 93% accuracy, Four-category classification	High accuracy with multiple CNN architectures
Varma et al. [65]	Fundus images	DCNN	92.7% accuracy, Efficient cataract diagnosis	Efficient cataract diagnosis with DCNN
Zhang et al. [57]	Fundus images	Multi-feature fusion, SVM, FCNN	92.66% average accuracy, Novel grading approach	High accuracy with novel grading approach
Chauhan et al. [72]	Fundus images	Ensemble (VGG-19, ResNet101V2, InceptionV3)	95.90% F-1 Score, Efficient cataract diagnosis	High F-1 Score with ensemble learning
Jones et al. [69]	66 images (normal, severe, mild)	DCNN (GoogLeNet architecture)	86.9% training accuracy, 35.8% validation accuracy	Significant drop in validation accuracy
Pratap et al. [70]	Fundus retinal images	Transfer learning with unspecified architecture	92.91% accuracy, Image quality selection module	High accuracy with quality selection module
Sahayam et al. [71]	ODIR dataset	EfficientNet-based CNN	99.21% sensitivity, 0.9921 accuracy, Automated diagnosis	High sensitivity and accuracy with automated diagnosis
Manikandan et al. [67]	Kaggle cataract dataset	GDCNN (Generative Adversarial Network, DCNN)	99.21% accuracy, Addressing cataract-related blindness	High accuracy, focus on blindness prevention
Khan et al. [62]	1,130 fundus images	CataractNet (custom deep neural network)	99.13% accuracy, Reduced computational cost	Very high accuracy with reduced computational cost
Faizal et al. [64]	Visible wavelength and anterior segment images	CNN (Inception-v3)	95% classification accuracy, Cost-effective solution	High accuracy and cost-effective approach
Yu et al. [54]	100 cataract procedures	SVM, RNN, CNN, CNN-RNN combinations	High unweighted accuracy, image-only CNN-RNN outperforms	High unweighted accuracy with CNN-RNN
Yusuf et al. [56]	100 eye images (normal, cataract)	Transfer learning with unspecified architecture	78% accuracy, Web-based CAD system	Web-based CAD system with moderate accuracy
Shetty et al. [61]	Not specified	VGG16	Addressing limitations, UI for cataract detection	Focus on UI for cataract detection
Arslan et al. [63]	2,748 Retinal Fundus images	DenseNet, EfficientNet, Xception, VGG, ResNet	94.88% accuracy, EfficientNet outperforms	EfficientNet achieved the highest accuracy
Angeline et al. [73]	Smartphone-captured eye images	Pretrained VGG16 model	92.1% accuracy, Early detection using smartphone images	Early detection with smartphone-captured images
Sahayam et al. [71]	ODIR dataset	EfficientNet-based CNN	99.21% sensitivity, 0.9921 accuracy, Automated diagnosis	High sensitivity and accuracy with automated diagnosis

thorough validation across populations remains essential to address these limitations and to ensure the widespread applicability and reliability of automated cataract detection systems.

D. SUMMARY

This section highlights the significant advancements in cataract detection facilitated by diverse machine-learning and deep-learning approaches. These methodologies, ranging from CNNs to ensemble techniques, demonstrate promising capabilities in the automated classification, grading, and diagnosis of cataracts using various datasets and innovative models. The high accuracies reported across different studies underscore the potential of these techniques to provide efficient and timely solutions for cataract detection. While acknowledging these achievements, it is essential to recognize the variability in the datasets, methodologies, and evaluation metrics employed, emphasizing the need for standardization in future research. Moreover, addressing

the limited sample sizes, potential biases, and necessity for real-world clinical validation remains crucial for the successful translation of these technologies into practical and widespread applications in the field of ophthalmology.

VI. AMBLYOPIA

Amblyopia, commonly known as “lazy eye,” is a visual developmental disorder characterized by reduced vision in one eye, which glasses or contact lenses cannot adequately correct. Typically occurring during early childhood, amblyopia results from a disruption in the normal visual development process, often due to factors such as strabismus (misalignment of the eyes), significant differences in refractive errors between the eyes, or visual deprivation, in which one eye experiences a lack of clear visual input. In response to conflicting visual signals from each eye during the critical period of visual development, the brain suppresses or prioritizes inputs from the stronger eye, leading to diminished visual acuity in the affected eye. Early detection

and intervention are crucial in the treatment of amblyopia, as the condition becomes more challenging to address with age. Amblyopia is a significant public health concern, emphasizing the importance of regular eye examinations in pediatric care to promptly identify and manage this condition.

A. METHODOLOGY AND DATASET VARIABILITY

When discussing amblyopia, the methodology and variety of the dataset are two important factors that affect how machine learning models are created and evaluated. Different research methods have been used in various studies. These include CNNs, ensemble techniques, and transfer learning. Some studies have focused on certain features or imaging methods, such as retinal images or patterns of visual stimuli, to improve the models ability to detect amblyopia. Different datasets have different sizes, makeups, and features. For example, some studies used datasets of visual stimuli presentations, whereas others used retinal images. This variety of methods and datasets shows how hard it is to use machine learning to treat amblyopia. This also shows the importance of using standardized methods to ensure that the results of testing different models for finding and treating amblyopia are consistent and reliable.

Marc Bosch et al. did a study to look into amblyopia, a condition that affects 2% to 3% of children. Researchers have examined neural network architectures as general feature extractors for two tasks: sorting eye images to identify strabismus and determining who needs to be referred to a specialist. Researchers looked into Several cutting-edge backbone architectures have been investigated. They found that VGG19 combined with a random forest classifier performed best for both tasks. Research has shown that advanced neural network architectures can help identify and fix problems related to amblyopia in young children [74]. Praveen et al. used image processing and machine learning to assist children with amblyopia. The method included taking pictures, using Canny edge detection to find people with amblyopia, and confirming the results with KNN, Logistic Regression, and the Random Forest Classifier (RFC). The use of Spyder IDE as the development environment, Python as the programming language, and OpenCV as a library for image processing are instrumental in implementing and testing these ML algorithms. It shown that, RFC is the best for detection of amblyopia [75]. Murali et al. mostly used an Android smartphone to create a quick and cheap way to check kids for amblyopia risk factors (ARF). For this study, pictures were taken under different lighting conditions, and deep learning and image processing algorithms were used to look at facial features linked to ARF. The model that was developed showed promise in its screening abilities, showing how smartphone-based photography could be used to find ARF and make referrals happen quickly [76]. Lalitha et al. used machine learning to improve the accuracy of finding amblyopia. The proposed method used RNNs to separate parts of an eye image, and used the Hausdorff Distance and Dice Coefficient similarity comparison to test for amblyopia.

When tested on 100 amblyopic images, the framework performed better than the other methods, indicating that it is useful for processing biomedical images [77]. Fan et al. aimed to improve the criteria for diagnosing amblyopia by looking at dynamic vision, especially eye movement, in children with amblyopia. The researchers devised a way to improve objects using a video eye tracker and an “artificial eye” to measure how the eyes move. A deep learning method was used to examine eye movement data for diagnostic and predictive purposes. The results showed significant differences in how the eyes moved between people with amblyopia and those with normal vision, which challenged what people previously thought. At the end of the study, certain eye movement parameters were suggested as ways to improve and add to the diagnostic criteria for amblyopia [78]. A prospective study by Arnaud Devlieger et al. compared the Blinq pediatric vision scanner to a standard eye exam (Gold Standard) to determine how well it could detect strabismus and amblyopia. The Blinq device worked well for detecting amblyopia and strabismus in 33.4% of the children, but it was not very good at detecting other eye problems. This study discussed the possible benefits of the bling device in detecting strabismus and poor foveolar fixation. This emphasized the sensitivity of the device for early detection of strabismus. However, it needs to be further improved before it can be used as a standalone tool for screening childrens eyesight [79]. Safa Ben Aoun et al. aimed to create deep learning models that could predict how children with open-globe eye injuries would perform with regard to their final visual acuity and amblyopia risk. Deep learning models such as neural networks, support vector machines (SVM), and decision trees were used on a dataset of 87 patients. The models were good at determining who would have poor vision and how likely they would develop amblyopia. They examined the initial vision, wound size, and anterior chamber inflammation. The results showed that predictive models can help identify high-risk groups and identify the best ways to monitor children who have had global injuries after surgery [80]. In 2023, Sengar et al. created an automated deep-learning framework for the noninvasive diagnosis of several eye diseases, focusing on early detection. EyeDeep-Net uses a keystone CNN to extract features from fundus images and performs better than baseline state-of-the-art models. Using digital fundus images [81], the framework offers a promising way to detect eye diseases early and treat them quickly.

Table 4 on amblyopia detection has showcased a range of methodologies, from advanced neural network architectures to smartphone-based applications, contributing to the ongoing efforts in early diagnosis and effective management of amblyopia-related issues. While some studies have explored the potential of deep learning models in predicting visual acuity prognosis and amblyopia risk, others have focused on optimizing diagnostic criteria through innovative approaches such as dynamic vision analysis. Despite these advancements, it is crucial to address these limitations and continue to refine

TABLE 4. Overview of studies on amblyopia detection.

Reference	Dataset Used	Adapted Methodology	Remarks	Key Findings
Devlieger et al. [79]	101 children aged 2 to 8 years	Assessed Blinq pediatric vision scanner compared to standard ophthalmic examination	Good sensitivity but limited specificity; Potential advantages in detecting poor foveolar fixation and strabismus, yet further improvement is needed for standalone use in pediatric vision screening	Effective in detecting foveolar fixation but needs improvement for standalone use
Aoun et al. [80]	Dataset of 87 patients with open globe eye injuries	Employed deep learning models for predicting final visual acuity prognosis and risk of amblyopia	Promising accuracy in predicting poor visual acuity prognosis and risk of amblyopia, highlighting significant indicators for each	High accuracy in predicting visual acuity and amblyopia risk
Lalitha et al. [77]	Dataset of 100 amblyopic images	RNN for Eye Image Segmentation	Proposed method demonstrated superior performance in amblyopia detection compared to existing methods	Superior performance in handling extensive datasets
Praveen et al. [75]	Dataset with parameters such as gender, age, cataract, myopia, hyperopia, strabismus, and class	Image processing and machine learning techniques using SPYDER IDE, OpenCV, and Python programming language	Showcased the potential of machine learning in healthcare applications for amblyopia detection	Demonstrates the use of machine learning in healthcare
Sengar et al. [81]	Multi-class eye disease RFMiD dataset	EyeDeep-Net with a keystone CNN	Introduced an automated deep learning-based framework for non-invasive diagnosis of multiple eye diseases	Superior performance compared to baseline models
Murali et al. [76]	Not specified	Smartphone-based photography with deep learning and image processing	Developed a cost-effective method for screening children for amblyopia risk factors using an Android smartphone	Promising screening capabilities for identifying risk factors
Fan et al. [78]	Not specified	Utilized video eye tracker combined with an "artificial eye"	Proposed optimization scheme for analyzing eye movement characteristics, challenging existing perceptions of amblyopic eyes, and suggesting specific eye movement parameters to enhance diagnostic criteria	New insights into eye movement parameters for diagnosis
Bosch et al. [74]	Not specified	Explored neural network architectures for feature extraction; VGG19 combined with a random forest classifier	VGG19 combined with a random forest classifier emerged as the top-performing combination for classifying eye images to detect strabismus and identifying the need for specialist referral	Top-performing combination for detecting strabismus and referral needs

these methods for widespread applicability in pediatric vision screening and healthcare.

B. MODEL ARCHITECTURE AND COMPARATIVE ANALYSIS

Studies on amblyopia detection have employed diverse model architectures, with the prevalent use of CNNs, such as VGG19, combined with a random forest classifier in the study by Bosch et al. Fan et al. leverage a deep learning approach without specifying the architecture, emphasizing the adaptability of deep learning in analyzing dynamic vision for enhanced diagnostic criteria. Lalitha et al. propose a method using RNNs for eye image segmentation, showcasing the efficacy of RNNs in handling sequential data for amblyopia detection. Sengar et al. introduce EyeDeep-Net, a keystone CNN, demonstrating superior performance in the non-invasive diagnosis of multiple eye diseases. The comparative analysis across these studies highlights the importance of tailoring model architectures to the unique challenges posed by amblyopia, emphasizing the need for specialized designs based on the nature of input data, and

encouraging further standardization and evaluation across diverse datasets to refine effective models for amblyopia detection.

C. LIMITATIONS

Despite advancements in machine learning applications for amblyopia detection, several limitations persist across studies. The lack of standardized datasets and methodologies poses a challenge, hindering direct comparisons between different models. The variability in dataset sizes, compositions, and characteristics makes it challenging to generalize the effectiveness of the proposed models across diverse populations. Additionally, some studies did not specify the datasets used, limiting their transparency and reproducibility. The reliance on specific features or imaging modalities in certain studies may overlook a broader spectrum of factors contributing to amblyopia. Furthermore, while deep learning models exhibit promising results, concerns regarding overfitting and generalizability to real-world clinical settings remain. Addressing these limitations is crucial for advancing

the reliability and applicability of machine learning models in amblyopia detection and for ensuring their seamless integration into pediatric vision screening and healthcare practices.

D. SUMMARY

In summary, the collective findings from diverse studies on amblyopia detection underscore the potential of machine learning applications, particularly leveraging CNNs, RNNs, and deep learning frameworks. These studies provide valuable insights into addressing amblyopia through innovative methodologies, ranging from smartphone-based photography to dynamic vision analysis. However, the field faces challenges related to dataset variability, lack of standardization, and concerns regarding generalizability to real-world clinical settings. These studies emphasize the need for ongoing refinement, standardization, and collaborative efforts to enhance the reliability and applicability of machine learning models for the early diagnosis and management of amblyopia. As the field progresses, addressing these limitations will be crucial for ensuring the seamless integration of machine learning technologies into pediatric vision screening and healthcare practices, ultimately improving the outcomes for children at risk of amblyopia-related issues.

VII. AGE-RELATED MACULAR DEGENERATION

Age-related macular degeneration (AMD) is a progressive and debilitating eye condition that primarily affects individuals aged 50 years and older and becomes more prevalent with advancing age. As the leading cause of irreversible vision loss in older people AMD poses a significant global public health challenge. This complex and multifactorial disease primarily targets the macula, a small but crucial area of the retina that is responsible for central vision. Deterioration of the macula leads to distorted or blurred vision, making tasks such as reading, recognizing faces, and driving increasingly difficult.

AMD exists in two main forms, dry (atrophic) and wet (exudative). Dry AMD, characterized by the gradual breakdown of light-sensitive cells in the macula, accounts for the majority of AMD cases. In contrast, wet AMD involves abnormal blood vessel growth beneath the macula, leading to leakage and potential scarring. Both forms can result in severe visual impairment, affecting the overall quality of life of the affected individuals.

A. METHODOLOGY AND DATASET VARIABILITY

With Age-Related Macular Degeneration (AMD), a full longitudinal observational design has been used to determine how the disease progresses in individuals aged ≥ 50 years. Case-control methods were employed to identify participants, ensuring a wide range of demographics and inclusion of different types of AMD such as dry and wet AMD.

Diagnostic criteria for accurate grouping of AMD cases included thorough eye examinations, including fundus photography and Optical Coherence Tomography (OCT) scans. Visual function tests, validated questionnaires, and interviews

were utilized to assess risk factors such as genetics, lifestyle, and medical history.

Follow-up assessments were performed regularly using consistent measurement tools to track changes in AMD status and visual function over time. Standardized imaging and measurement techniques ensured data consistency, while outliers and methodological differences were examined to understand their impact on study results.

Several studies have focused on developing deep learning models for AMD detection using OCT images. Kadry et al. proposed an automated system combining deep and hand-crafted features using VGG16 for feature extraction, achieving accuracies of 97.08% on fundus retinal images and 97.50% on OCT images [82]. Han et al. developed a CNN model to differentiate types of neovascular AMD using SD-OCT images, achieving 87.4% accuracy and outperforming ophthalmologists on the same dataset [84]. Ma et al. utilized a deep learning framework on volumetric OCT images to distinguish between Polypoidal Choroidal Vasculopathy (PCV) and wet AMD, achieving over 90% accuracy [85]. He T et al. proposed a method using a retrained ResNet-50 model with L2-constrained softmax loss and a Local Outlier Factor algorithm, achieving 99.87% accuracy on the UCSD dataset and 97.56% on the Duke dataset [86]. Motozawa et al. constructed two CNN models to automate OCT image interpretation in AMD, showing high sensitivity, specificity, and accuracy [87].

Transfer learning has been leveraged to improve model performance in AMD detection. Han et al. used transfer learning and data augmentation to enhance the robustness of their CNN model [84]. Choudhary et al. employed transfer learning on a large set of retinal OCT images using a custom 19-layer DCNN based on the VGG-19 architecture, outperforming other models in automatic retinal disease identification [94]. Mishra et al. incorporated a deformation-aware attention mechanism into a deep attention-based CNN model and used transfer learning with limited training data, achieving state-of-the-art performance on four datasets [98].

Systematic reviews have highlighted progress and challenges in using deep learning for AMD detection. Treder et al. conducted a systematic review on the application of deep learning in OCT for AMD, identifying key topics such as classification, segmentation, and treatment response assessment, and emphasizing the need for prospective studies and external validations [83]. Renu Deepti et al. reviewed 50 papers focusing on AI techniques in AMD detection, particularly highlighting the superiority of deep learning models over other methods and suggesting further research to address existing gaps [97].

Novel architectures and high-performance models have been proposed to enhance AMD detection accuracy. Ali et al. introduced the hybrid deep AMDNet23 model combining CNNs with LSTM systems, achieving an accuracy of 96.50% and outperforming 13 pre-trained CNN models [96]. Choudhary et al. demonstrated the effectiveness of their

custom 19-layer DCNN based on the VGG-19 architecture in classifying retinal diseases [94]. Mishra et al. showcased their advanced computational tool incorporating attention mechanisms, achieving high diagnostic accuracy with fewer network parameters [98].

Efforts in automating retinal layer segmentation in OCT images include the work of Mukherjee et al., who proposed a method utilizing three-dimensional spatial context through a semantic segmentation algorithm. This approach preserves continuity and anatomical relationships, performing well on an AMD dataset and highlighting the benefits of considering the entire OCT volume [93].

Studies focusing on early detection and the need for automated tools have been conducted by Chakraborty et al., who proposed a novel deep-learning model capable of self-learning and feature determination, achieving an 84% success rate in detecting early and intermediate stages of asymptomatic AMD [90]. Udayaraju et al. discussed the use of deep learning to analyze OCT images of dry and wet ARMD, evaluating different preprocessing and feature extraction methods to improve early diagnosis [91]. Chakraborty et al. also proposed a new 13-layer DCNN structure for fundus image analysis, enhancing the accuracy of AMD detection [92].

Preprocessing techniques have been explored to improve diagnostic systems. Ali et al. aimed to simplify eye disease diagnosis by proposing a computer system that uses Contrast Limited Adaptive Histogram Equalization (CLAHE) and gamma correction, combining CNNs with LSTM systems. Their hybrid deep AMDNet23 model achieved impressive accuracy, demonstrating its potential as a state-of-the-art diagnostic tool [96]. Prasad et al. addressed ARMD detection by using Ant Colony Optimization for macular region segmentation and SVM for classification, achieving high specificity, accuracy, and sensitivity [89].

Advancements in automated grading and risk assessment have been made by Peng et al., who developed DeepSeeNet, a deep-learning model that automates classification based on the AREDS Simplified Severity Scale. This model outperformed retinal specialists and aids in early AMD detection and risk assessment [88].

Finally, comprehensive screening systems integrating hardware and software innovations have been proposed. Selvakumar et al. created an eye-screening system using OCT and deep learning, implementing the DLCTO-AMDC model with Inception v3 as the feature extractor. Their work suggests the potential of wearable devices for early AMD detection [95].

These diverse studies in Table 5 highlights the remarkable strides made in leveraging advanced technologies, particularly deep learning, for the early and accurate detection of Age-Related Macular Degeneration. With innovative methodologies and promising results, these approaches not only enhance our understanding of AMD but also hold substantial potential for transforming clinical practices, offering efficient diagnostic tools to address the

challenges posed by this prevalent and impactful ocular condition.

B. MODEL ARCHITECTURE AND COMPARATIVE ANALYSIS

Various studies have employed deep-learning techniques, often utilizing CNNs such as VGG16, VGG19, ResNet, and custom architectures. A comparative analysis across studies indicated that different models were selected based on the specific objectives of each study, such as detecting AMD, distinguishing its subtypes, or automating retinal layer segmentation.

The studies consistently demonstrated high accuracy in AMD detection, showing the effectiveness of deep learning models in analyzing diverse imaging modalities, including fundus retinal images (FRI) and OCT images. Several studies have employed transfer learning, data augmentation, and innovative preprocessing techniques to enhance the robustness and performance of the model. Despite variations in datasets and methodologies, a consistent theme is the application of deep learning to address the challenges of AMD diagnosis. Comparative analysis suggests that these models, with their diverse architectures and approaches, have the potential to contribute significantly to the field of ophthalmology by offering accurate and efficient tools for AMD detection and classification.

C. LIMITATIONS

One notable limitation of these studies is the potential lack of standardization of datasets, preprocessing techniques, and evaluation metrics. The diversity of datasets, encompassing various imaging modalities and acquisition devices, may introduce heterogeneity, affecting the generalizability of the developed models. Differences in the preprocessing steps, feature extraction methods, and model architectures make direct comparisons challenging. Standardized benchmarks and protocols for data pre-processing, model development, and evaluation metrics are essential for establishing consistent and reliable performance assessments. Moreover, the studies predominantly focused on binary classification tasks related to AMD detection, and further research is needed to address the broader spectrum of AMD subtypes and disease progression, ensuring comprehensive and clinically relevant applications.

D. SUMMARY

This section describes the remarkable strides made by leveraging deep learning techniques to detect and classify Age-Related Macular Degeneration (AMD). The diverse methodologies employed, ranging from intricate feature extraction and fusion to advanced CNNs, show the versatility of artificial intelligence in ophthalmic image analysis. While achieving notable accuracy and outperforming traditional methods, the field faces challenges in standardization across datasets, preprocessing techniques, and evaluation metrics. The potential for automation and efficiency in AMD diagnosis is evident, emphasizing the significance of ongoing

TABLE 5. Summary of studies on age-related macular degeneration (AMD) detection.

Reference	Dataset Used	Adapted Methodology	Remarks	Key Findings
He et al. [86]	OCT images (UCSD and Duke datasets)	Retrained ResNet-50 with L2-constrained softmax loss for feature extraction; LOF algorithm as classifier	Achieved high accuracy (99.87% on UCSD, 97.56% on Duke), demonstrating robust performance across datasets	Outstanding accuracy across multiple datasets
Ma et al. [85]	Volumetric OCT images	Single-B-scan-based classification, volumetric probability prediction aggregation, comparison of labeling strategies	Demonstrated over 90% differentiation accuracy between PCV and wet AMD, with good correspondence to ICGA-confirmed pathologies	Effective differentiation between AMD types
Mukherjee et al. [93]	OCT images (AMD dataset)	Algorithm for retinal layer segmentation using 3D spatial context	Demonstrated favorable performance in an AMD dataset, emphasizing the utility of treating the entire OCT volume as a correlated entity	Effective segmentation with 3D context
Motozawa et al. [87]	OCT images	Development of CNN models for AMD classification and exudative changes; Visualization through class activation mapping (CAM)	Achieved high sensitivity, specificity, and accuracy, showcasing potential for automating OCT image interpretation	High performance in automated image interpretation
Udayaraju et al. [91]	OCT images	Utilized deep learning to analyze OCT images of dry and wet ARMD effectively	Focused on addressing the complexity of retinal diseases, particularly ARMD, aiming to contribute to improved understanding and management	Improved understanding and management of ARMD
Choudhary et al. [94] (2023)	OCT retinal images	VGG-19 with transfer learning for automatic retinal disease detection	Achieved impressive classification accuracy, surpassing existing models	High accuracy with VGG-19 and transfer learning
Selvakumar et al. [95] (2023)	OCT images	DLCTO-AMDC model for early detection and classification of AMD	Demonstrated superiority in classification accuracy, providing a promising solution for early detection	Superior early detection and classification
Kadry et al. [82]	Not specified	Initial data processing, VGG16 for deep feature extraction, handcrafted features (LBP, PHOG, DWT), Mayfly algorithm for feature selection	Achieved high accuracies with the proposed VGG16 and concatenated features for AMD detection in FRI and OCT images	High accuracy with VGG16 and concatenated features
Han et al. [84]	SD-OCT images of nAMD patients and controls	CNN (VGG-16, VGG-19, ResNet) training and validation for distinguishing nAMD subtypes; Transfer learning and data augmentation	Achieved 87.4% accuracy on the test data, outperforming ophthalmologists on the same dataset	Outperformed ophthalmologists in nAMD subtype classification
Peng et al. [88]	Bilateral color fundus photographs	Development of DeepSeeNet for patient-based classification using AREDS Simplified Severity Scale	Outperformed retinal specialists in patient-based classification, suggesting potential to aid clinical decision-making for AMD severity	High performance in patient-based classification
Prasad et al. [89]	Fundus images	Automated system using ACO for macular region segmentation and SVM for classification	Demonstrated high specificity, accuracy, and sensitivity, surpassing previous methods in the literature	High specificity and sensitivity
Chakraborty et al. [90]	Retinal images	Novel deep learning model for AMD detection; Detection accuracy of 84%	Effective screening of retinal images for direct evidence of AMD	Detection accuracy of 84% with novel deep learning model
Chakraborty et al. [92]	Fundus images (iChallenge-AMD and ARIA)	Novel 13-layer DCNN for screening fundus images to identify direct signs of AMD	Outperformed the best-known algorithm, showcasing its effectiveness in enhancing AMD detection accuracy	Outperformed existing algorithms
Treder et al. [83]	SD-OCT images	Pretrained DCNN for automated AMD detection; Testing and validation of DCNN's performance	Demonstrated high sensitivity and specificity, suggesting its potential as a supportive tool in clinical decisions for AMD detection	High sensitivity and specificity
Ali et al. [96]	Fundus ophthalmology images	Framework with CLAHE, Gamma correction, CNNs, and LSTM; Evaluation of hybrid deep AMDNet23 model	Achieved a remarkable accuracy of 96.50%, outperforming pre-trained CNN models	High accuracy with hybrid deep learning model
Mishra et al. [98]	OCT scans	Attention-based deep CNN model with transfer learning	Achieved state-of-the-art performance across datasets, emphasizing efficiency with reduced network parameters	State-of-the-art performance with reduced parameters

research on refining methodologies, ensuring generalizability, and addressing the complexities of AMD subtypes and disease progression. The integration of artificial intelligence in AMD detection holds promise for improved clinical workflows, early intervention, and enhanced outcomes in managing this prevalent cause of visual impairment.

VIII. CONCLUSION

This comprehensive review underscores the significant advances made in the automated diagnosis of ophthalmic diseases using deep learning techniques. The amalgamation of 99 recent studies has provided a panoramic view of the transformative impact of deep learning methodologies, such as CNNs, RNNs, and hybrid models, which have been used to enhance the accuracy, efficiency, and accessibility of disease detection in the field of ophthalmology.

Exploring diverse imaging modalities, including fundus photography, OCT, and visual field testing, has further expanded the scope and versatility of deep-learning applications. Each disease category, encompassing glaucoma, diabetic retinopathy, cataract, amblyopia, and macular degeneration, has been meticulously scrutinized, highlighting the unique challenges and promising opportunities associated with automated diagnosis.

The key findings and advancements highlighted in this review underscore the potential of deep learning algorithms in facilitating early disease detection and enabling timely intervention. The prospect of revolutionizing ophthalmic diagnostics through these advancements holds great promise for improving patient outcomes and reducing the burden of preventable vision impairments. However, it is crucial to acknowledge existing limitations, including data variability, interpretability, and the necessity for large, diverse datasets to ensure the robustness and generalizability of deep learning models. As the field progresses, addressing these challenges is paramount for the continued development and deployment of reliable and efficient deep-learning-based diagnostic tools for eye diseases.

Given these insights, this literature synthesis will guide future research. The identified gaps and challenges provide a roadmap for researchers to navigate, foster the evolution of innovative solutions, and further refine deep learning algorithms in ophthalmic diagnostics. By collaboratively addressing these challenges, the scientific community can pave the way for integrating advanced technologies into clinical practice, ultimately improving the accessibility and effectiveness of automated diagnosis of ophthalmic diseases.

A collective group of authors acknowledged that data scarcity is a critical challenge in the development of deep learning models for ophthalmic disease diagnosis. Indeed, the integration of real-time data acquisition systems in live clinical environments offers a promising solution to this limitation. By implementing such systems, data can be continuously captured and fed into the model, allowing for ongoing learning and refinement of diagnostic algorithms. This approach would address the need for larger and more diverse datasets, helping models to adapt to a broader spectrum of real-world variability, including different patient demographics, disease stages, and imaging modalities.

Moreover, leveraging real-time data acquisition could facilitate quicker model updates and enable adaptive learning in response to emerging patterns, thereby improving the

system's diagnostic accuracy and robustness. To ensure patient privacy and data security, federated learning could be explored, allowing multiple clinical institutions to contribute data without compromising patient confidentiality. Such collaborative efforts would significantly enhance the generalizability of the models while adhering to ethical standards in medical research.

REFERENCES

- [1] S. Singh, N. Sambyal, and A. Aggarwal, "Automated glaucoma detection using deep convolutional neural networks," *Res. Square*, Apr. 2023, doi: [10.21203/rs.3.rs-2788554/v1](https://doi.org/10.21203/rs.3.rs-2788554/v1). [Online]. Available: <https://www.researchsquare.com/article/rs-2788554/v1>
- [2] R. Sathya and P. Balamurugan, "Glaucoma identification in digital fundus images using deep learning enhanced auto encoder networks (DL-EAEN) for accurate diagnosis," *Indian J. Sci. Technol.*, vol. 16, no. 44, pp. 4026–4037, Nov. 2023, doi: [10.17485/ijst/v16i44.2325](https://doi.org/10.17485/ijst/v16i44.2325).
- [3] T. Maurya, L. Kala, K. Manasa, and K. Gunasekaran, "Retinal glaucoma detection using deep learning algorithm," *Int. J. Intell. Syst. Appl. Eng.*, vol. 10, no. 1, pp. 52–59, Mar. 2022, doi: [10.18201/ijisae.2022.267](https://doi.org/10.18201/ijisae.2022.267).
- [4] M. B. Sudhan, M. Sinthuja, S. P. Raja, J. Amutharaj, G. C. P. Latha, S. S. Rachel, T. Anitha, T. Rajendran, and Y. A. Waji, "Segmentation and classification of glaucoma using U-Net with deep learning model," *J. Healthcare Eng.*, vol. 2022, pp. 1–10, Feb. 2022. [Online]. Available: <https://www.hindawi.com/journals/jhe/2022/1601354/>
- [5] R. Kashyap, R. Nair, S. M. P. Gangadharan, M. Botto-Tobar, S. Farooq, and A. Rizwan, "Glaucoma detection and classification using improved U-Net deep learning model," *Healthcare*, vol. 10, no. 12, p. 2497, Dec. 2022, doi: [10.3390/healthcare10122497](https://doi.org/10.3390/healthcare10122497).
- [6] S. I. Khan, S. B. Choubey, A. Choubey, A. Bhatt, P. V. Naishadhkumar, and M. M. Basha, "Automated glaucoma detection from fundus images using wavelet-based denoising and machine learning," *Concurrent Eng.*, vol. 30, no. 1, pp. 103–115, Mar. 2022, doi: [10.1177/1063293x211026620](https://doi.org/10.1177/1063293x211026620).
- [7] S. Ajitha, J. D. Akkara, and M. V. Judy, "Identification of glaucoma from fundus images using deep learning techniques," *Indian J. Ophthalmol.*, vol. 69, no. 10, pp. 2702–2709, Oct. 2021, doi: [10.4103/ijo.ijo_92_21](https://doi.org/10.4103/ijo.ijo_92_21).
- [8] S. S. Vanjire, "Glaucoma detection using deep learning," *Int. J. Res. Publication Rev.*, vol. 2, no. 8, pp. 1330–1335, 2021.
- [9] P. M. S. Raja and S. L. Jothilakshmi, "Deep learning algorithms and glaucoma detection: A review," *Int. Res. J. Eng. Technol.*, vol. 8, no. 2, pp. 1185–1196, 2021.
- [10] M. Juneja, N. Thakur, S. Thakur, A. Uniyal, A. Wani, and P. Jindal, "GC-NET for classification of glaucoma in the retinal fundus image," *Mach. Vis. Appl.*, vol. 31, no. 5, pp. 1–18, Jul. 2020, doi: [10.1007/s00138-020-01091-4](https://doi.org/10.1007/s00138-020-01091-4).
- [11] R. Janani and S. P. Rajamohana, "Early detection of glaucoma using optic disc and optic cup segmentation: A survey," *Mater. Today, Proc.*, vol. 45, pp. 2763–2769, Jan. 2021.
- [12] S. Chandrappa, L. Dharmanna, B. Anami, C. Nandini, and M. Rafi, "A novel approach to glaucoma screening using optic nerve head through image fusion and fractal geometry," *J. Math. Comput. Sci.*, vol. 11, no. 4, pp. 4284–4300, 2021.
- [13] C. Sharmila and N. Shanthi, "Retinal image analysis for glaucoma detection using transfer learning," in *Advances in Electrical and Computer Technologies (Lecture Notes in Electrical Engineering)*, vol. 711, T. Sengodan, M. Murugappan, and S. Misra, Eds., Singapore: Springer, 2021, doi: [10.1007/978-981-15-9019-1_21](https://doi.org/10.1007/978-981-15-9019-1_21).
- [14] S. S. Kanse and D. M. Yadav, "HG-SVNN: Harmonic genetic-based support vector neural network classifier for the glaucoma detection," *J. Mech. Med. Biol.*, vol. 20, no. 1, Feb. 2020, Art. no. 1950065.
- [15] R. Wang, C. Bradley, P. Herbert, K. Hou, P. Ramulu, K. Breininger, M. Unberath, and J. Yohannan, "Deep learning-based identification of eyes at risk for glaucoma surgery," *Sci. Rep.*, vol. 14, no. 1, p. 599, Jan. 2024, doi: [10.1038/s41598-023-50597-0](https://doi.org/10.1038/s41598-023-50597-0).
- [16] M. Ashtari-Majlan, M. M. Dhehibi, and D. Masip, "Deep learning and computer vision for glaucoma detection: A review," 2023, *arXiv:2307.16528*.
- [17] A. Shoukat, S. Akbar, S. A. Hassan, S. Iqbal, A. Mehmood, and Q. M. Ilyas, "Automatic diagnosis of glaucoma from retinal images using deep learning approach," *Diagnostics*, vol. 13, no. 10, p. 1738, May 2023, doi: [10.3390/diagnostics13101738](https://doi.org/10.3390/diagnostics13101738).

- [18] S. Saha, J. Vignarajan, and S. Frost, "A fast and fully automated system for glaucoma detection using color fundus photographs," *Sci. Rep.*, vol. 13, no. 1, p. 18408, Oct. 2023, doi: [10.1038/s41598-023-44473-0](https://doi.org/10.1038/s41598-023-44473-0).
- [19] N. S. Shadin, S. Sanjana, S. Chakraborty, and N. Sharmin, "Performance analysis of glaucoma detection using deep learning models," in *Proc. Int. Conf. Innov. Sci., Eng. Technol.*, 2022, pp. 190–195, doi: [10.1109/ICISSET54810.2022.9775828](https://doi.org/10.1109/ICISSET54810.2022.9775828).
- [20] E. Noury, S. S. Mannil, R. T. Chang, A. R. Ran, C. Y. Cheung, S. S. Thapa, H. L. Rao, S. Dasari, M. Riyazuddin, D. Chang, S. Nagaraj, C. C. Tham, and R. Zadeh, "Deep learning for glaucoma detection and identification of novel diagnostic areas in diverse real-world datasets," *Transl. Vis. Sci. Technol.*, vol. 11, no. 5, p. 11, May 2022, doi: [10.1167/tvst.11.5.11](https://doi.org/10.1167/tvst.11.5.11).
- [21] N. Akter, J. Fletcher, S. Perry, M. P. Simunovic, N. Briggs, and M. Roy, "Glaucoma diagnosis using multi-feature analysis and a deep learning technique," *Sci. Rep.*, vol. 12, no. 1, p. 8064, May 2022, doi: [10.1038/s41598-022-12147-y](https://doi.org/10.1038/s41598-022-12147-y).
- [22] R. Hemelings, B. Elen, J. Barbosa-Breda, M. B. Blaschko, P. De Boever, and I. Stalmans, "Deep learning on fundus images detects glaucoma beyond the optic disc," *Sci. Rep.*, vol. 11, no. 1, p. 20313, Oct. 2021, doi: [10.1038/s41598-021-99605-1](https://doi.org/10.1038/s41598-021-99605-1).
- [23] G. D'Souza, P. C. Siddalingaswamy, and M. A. Pandya, "AlterNet-K: A small and compact model for the detection of glaucoma," *Biomed. Eng. Lett.*, vol. 14, no. 1, pp. 23–33, Jan. 2024, doi: [10.1007/s13534-023-00307-6](https://doi.org/10.1007/s13534-023-00307-6).
- [24] P. Parkhi and B. Hambarde, "Optical cup and disc segmentation using deep learning technique for glaucoma detection," *Int. J. Next-Gener. Comput.*, vol. 14, no. 1, pp. 44–52, 2023, doi: [10.47164/ijngc.v14i1.1017](https://doi.org/10.47164/ijngc.v14i1.1017).
- [25] R. Mahum, S. U. Rehman, O. D. Okon, A. Alabrah, T. Meraj, and H. T. Rauf, "A novel hybrid approach based on deep CNN to detect glaucoma using fundus imaging," *Electronics*, vol. 11, no. 1, p. 26, Dec. 2021, doi: [10.3390/electronics11010026](https://doi.org/10.3390/electronics11010026).
- [26] M. Christopher, A. Belghith, C. Bowd, J. A. Proudfoot, M. H. Goldbaum, R. N. Weinreb, C. A. Girkin, J. M. Liebmann, and L. M. Zangwill, "Performance of deep learning architectures and transfer learning for detecting glaucomatous optic neuropathy in fundus photographs," *Sci. Rep.*, vol. 8, no. 1, p. 16685, Nov. 2018, doi: [10.1038/s41598-018-35044-9](https://doi.org/10.1038/s41598-018-35044-9).
- [27] J. Lee, Y. W. Kim, A. Ha, Y. K. Kim, K. H. Park, H. J. Choi, and J. W. Jeoung, "Estimating visual field loss from monoscopic optic disc photography using deep learning model," *Sci. Rep.*, vol. 10, no. 1, p. 21052, Dec. 2020, doi: [10.1038/s41598-020-78144-1](https://doi.org/10.1038/s41598-020-78144-1).
- [28] A. A. Jammal, A. C. Thompson, N. G. Ogata, E. B. Mariottoni, C. N. Urata, V. P. Costa, and F. A. Medeiros, "Detecting retinal nerve fibre layer segmentation errors on spectral domain-optical coherence tomography with a deep learning algorithm," *Sci. Rep.*, vol. 9, no. 1, p. 9836, Jul. 2019.
- [29] K. Park, J. Kim, and J. Lee, "Visual field prediction using recurrent neural network," *Sci. Rep.*, vol. 9, no. 1, p. 8385, Jun. 2019, doi: [10.1038/s41598-019-44852-6](https://doi.org/10.1038/s41598-019-44852-6).
- [30] S. Mishra, S. Hanchate, and Z. Saquib, "Diabetic retinopathy detection using deep learning," in *Proc. Int. Conf. Smart Technol. Comput., Electr. Electron. (ICSTCEE)*, Bengaluru, India, Oct. 2020, pp. 515–520, doi: [10.1109/ICSTCEE49637.2020.9277506](https://doi.org/10.1109/ICSTCEE49637.2020.9277506).
- [31] P. Saranya, R. Pranati, and S. S. Patro, "Detection and classification of red lesions from retinal images for diabetic retinopathy detection using deep learning models," *Multimedia Tools Appl.*, vol. 82, no. 25, pp. 39327–39347, Oct. 2023, doi: [10.1007/s11042-023-15045-1](https://doi.org/10.1007/s11042-023-15045-1).
- [32] G. Kalyani, B. Janakiramaiah, A. Karuna, and L. V. N. Prasad, "Diabetic retinopathy detection and classification using capsule networks," *Complex Intell. Syst.*, vol. 9, no. 3, pp. 2651–2664, Jun. 2023, doi: [10.1007/s40747-021-00318-9](https://doi.org/10.1007/s40747-021-00318-9).
- [33] S. Bhandari, S. Pathak, and S. A. Jain, "A literature review of early-stage diabetic retinopathy detection using deep learning and evolutionary computing techniques," *Arch. Comput. Methods Eng.*, vol. 30, no. 2, pp. 799–810, Mar. 2023, doi: [10.1007/s11831-022-09816-6](https://doi.org/10.1007/s11831-022-09816-6).
- [34] D. Das, S. K. Biswas, and S. Bandyopadhyay, "Detection of diabetic retinopathy using convolutional neural networks for feature extraction and classification (DRFEC)," *Multimedia Tools Appl.*, vol. 82, no. 19, pp. 29943–30001, Aug. 2023, doi: [10.1007/s11042-022-14165-4](https://doi.org/10.1007/s11042-022-14165-4).
- [35] C. Mohanty, S. Mahapatra, B. Acharya, F. Kokkoras, V. C. Gerogiannis, I. Karamitsos, and A. Kanavos, "Using deep learning architectures for detection and classification of diabetic retinopathy," *Sensors*, vol. 23, no. 12, p. 5726, Jun. 2023, doi: [10.3390/s23125726](https://doi.org/10.3390/s23125726).
- [36] G. Sivapriya, R. Manjula Devi, P. Keerthika, and V. Praveen, "Automated diagnostic classification of diabetic retinopathy with microvascular structure of fundus images using deep learning method," *Biomed. Signal Process. Control*, vol. 88, Feb. 2024, Art. no. 105616.
- [37] L. G. Atlas, K. P. Arjun, K. S. Kumar, R. K. Dhanaraj, and A. Nayyar, "Deep learning enabled hemorrhage detection in retina with DPFE and splat segmentation in fundus images," *Biomed. Signal Process. Control*, vol. 88, Feb. 2024, Art. no. 105692.
- [38] A. K. Gangwar and V. Ravi, "Diabetic retinopathy detection using transfer learning and deep learning," in *Evolution in Computational Intelligence (Advances in Intelligent Systems and Computing)*, vol. 1176, V. Bhateja, S. L. Peng, S. C. Satapathy, and Y. D. Zhang, Eds., Singapore: Springer, 2021, doi: [10.1007/978-981-15-5788-0_64](https://doi.org/10.1007/978-981-15-5788-0_64).
- [39] D. Doshi, A. Shenoy, D. Sidhpura, and P. Gharpure, "Diabetic retinopathy detection using deep convolutional neural networks," in *Proc. Int. Conf. Comput., Analytics Secur. Trends (CAST)*, Pune, India, Dec. 2016, pp. 261–266, doi: [10.1109/CAST.2016.7914977](https://doi.org/10.1109/CAST.2016.7914977).
- [40] D. J. Hemanth, O. Deperlioglu, and U. Kose, "An enhanced diabetic retinopathy detection and classification approach using deep convolutional neural network," *Neural Comput. Appl.*, vol. 32, no. 3, pp. 707–721, Feb. 2020, doi: [10.1007/s00521-018-03974-0](https://doi.org/10.1007/s00521-018-03974-0).
- [41] K. N. S. Sree, D. V. Sree, G. H. Lakshmi, and S. Ramesh, "Diabetic retinopathy detection using deep learning," in *Proc. 2nd Int. Conf. Electron. Sustain. Commun. Syst. (ICESC)*, Aug. 2021, pp. 1670–1674.
- [42] T. M. Usman, Y. K. Saheed, D. Ignace, and A. Nsang, "Diabetic retinopathy detection using principal component analysis multi-label feature extraction and classification," *Int. J. Cognit. Comput. Eng.*, vol. 4, pp. 78–88, Jun. 2023.
- [43] I. Qureshi, J. Ma, and Q. Abbas, "Diabetic retinopathy detection and stage classification in eye fundus images using active deep learning," *Multimedia Tools Appl.*, vol. 80, no. 8, pp. 11691–11721, 2021, doi: [10.1007/s11042-020-10238-4](https://doi.org/10.1007/s11042-020-10238-4).
- [44] M. Chetoui, M. A. Akhloufi, and M. Kardouchi, "Diabetic retinopathy detection using machine learning and texture features," in *Proc. IEEE Can. Conf. Electr. Comput. Eng. (CCECE)*, May 2018, pp. 1–4, doi: [10.1109/CCECE.2018.8447809](https://doi.org/10.1109/CCECE.2018.8447809).
- [45] S. Muhammad Saiful Islam, M. Mahedi Hasan, and S. Abdullah, "Deep learning based early detection and grading of diabetic retinopathy using retinal fundus images," 2018, *arXiv:1812.10595*.
- [46] E. Özbay, "An active deep learning method for diabetic retinopathy detection in segmented fundus images using artificial bee colony algorithm," *Artif. Intell. Rev.*, vol. 56, pp. 3291–3318 Apr. 2023, doi: [10.1007/s10462-022-10231-3](https://doi.org/10.1007/s10462-022-10231-3).
- [47] H. R. H. Al-Absi, A. Pai, U. Naeem, F. K. Mohamed, S. Arya, R. A. Sheit, M. Bashir, M. M. El Shafei, N. El Hajj, and T. Alam, "DiaNet v2 deep learning based method for diabetes diagnosis using retinal images," *Sci. Rep.*, vol. 14, no. 1, p. 1595, Jan. 2024, doi: [10.1038/s41598-023-49677-y](https://doi.org/10.1038/s41598-023-49677-y).
- [48] L. Dai, B. Sheng, T. Chen, Q. Wu, R. Liu, C. Cai, L. Wu, D. Yang, H. Hamzah, Y. Liu, and X. Wang, "A deep learning system for predicting time to progression of diabetic retinopathy," *Nature Med.*, vol. 30, no. 2, pp. 584–594, 2024, doi: [10.1038/s41591-023-02702-z](https://doi.org/10.1038/s41591-023-02702-z).
- [49] M. R. Shoaib, H. M. Emara, J. Zhao, W. El-Shafai, N. F. Soliman, A. S. Mubarak, O. A. Omer, F. E. A. El-Samie, and H. Esmail, "Deep learning innovations in diagnosing diabetic retinopathy: The potential of transfer learning and the DiaCNN model," *Comput. Biol. Med.*, vol. 169, Feb. 2024, Art. no. 107834.
- [50] A. A. Bosale, "Detection and classification of diabetic retinopathy using deep learning algorithms for segmentation to facilitate referral recommendation for test and treatment prediction," 2024, *arXiv:2401.02759*.
- [51] R. Al-Ahmadi, H. Al-Ghamdi, and L. Hsairi, "Classification of diabetic retinopathy by deep learning," *Int. J. Online Biomed. Eng.*, vol. 20, no. 1, pp. 74–88, 2024.
- [52] D. A. Hasan, S. R. M. Zeebaree, M. A. M. Sadeeq, H. M. Shukur, R. R. Zebari, and A. H. Alkhayyat, "Machine learning-based diabetic retinopathy early detection and classification systems—A survey," in *Proc. 1st Babylon Int. Conf. Inf. Technol. Sci. (BICITS)*, Babil, Iraq, Apr. 2021, pp. 16–21, doi: [10.1109/BICITS51482.2021.9509920](https://doi.org/10.1109/BICITS51482.2021.9509920).
- [53] M. M. Islam, H.-C. Yang, T. N. Poly, W.-S. Jian, and Y.-C. J. Li, "Deep learning algorithms for detection of diabetic retinopathy in retinal fundus photographs: A systematic review and meta-analysis," *Comput. Methods Programs Biomed.*, vol. 191, Jul. 2020, Art. no. 105320.

- [54] F. Yu, G. S. Croso, T. S. Kim, Z. Song, F. Parker, G. D. Hager, A. Reiter, S. S. Vedula, H. Ali, and S. Sikder, "Assessment of automated identification of phases in videos of cataract surgery using machine learning and deep learning techniques," *JAMA Netw. Open.*, vol. 2, no. 4, 2019, Art. no. e191860, doi: [10.1001/jamanetworkopen.2019.1860](https://doi.org/10.1001/jamanetworkopen.2019.1860).
- [55] J. Ran, K. Niu, Z. He, H. Zhang, and H. Song, "Cataract detection and grading based on combination of deep convolutional neural network and random forests," in *Proc. Int. Conf. Netw. Infrastructure Digit. Content (IC-NIDC)*, Guiyang, China, 2018, pp. 155–159, doi: [10.1109/ICNIDC.2018.8525852](https://doi.org/10.1109/ICNIDC.2018.8525852).
- [56] M. Yusuf, S. Theophilous, J. Adejoke, and A. B. Hassan, "Web-based cataract detection system using deep convolutional neural network," in *Proc. 2nd Int. Conf. IEEE Nigeria Comput. Chapter (NigeriaComputConf)*, Zaria, Nigeria, 2019, pp. 1–7, doi: [10.1109/NigeriaComputConf45974.2019.8949636](https://doi.org/10.1109/NigeriaComputConf45974.2019.8949636).
- [57] H. Zhang, K. Niu, Y. Xiong, W. Yang, Z. He, and H. Song, "Automatic cataract grading methods based on deep learning," *Comput. Methods Programs Biomed.*, vol. 182, Dec. 2019, Art. no. 104978.
- [58] R. B. J. Simanjuntak, Y. Fu'adah, R. Magdalena, S. Saidah, A. B. Wiratama, and I. D. S. Ubaidah, "Cataract classification based on fundus images using convolutional neural network," *JOIV : Int. J. Informat. Visualizat.*, vol. 6, no. 1, p. 33, Mar. 2022.
- [59] Y. Wang, C. Tang, J. Wang, Y. Sang, and J. Lv, "Cataract detection based on ocular B-ultrasound images by collaborative monitoring deep learning," *Knowledge-Based Syst.*, vol. 231, Nov. 2021, Art. no. 107442.
- [60] X. Huang, H. Mori, D. Lukmanto, T. H. Tran, M. Fukuda, T. Oshika, S. Yousefi, S. Fukuda, and J. Sun, "Objective cataract detection and grading with deep learning based on OCT densitometry," *Investigative Ophthalmol. Visual Sci.*, vol. 62, no. 11, p. 67, 2021.
- [61] A. Shetty, K. Bathija, and R. L. Priya, "Cataract eye detection using machine learning models," in *Emerging Technologies in Data Mining and Information Security (Lecture Notes in Networks and Systems)*, vol. 164, J. M. R. S. Tavares, S. Chakrabarti, A. Bhattacharya, and S. Ghatak, Eds., Singapore: Springer, 2021, doi: [10.1007/978-981-15-9774-9_84](https://doi.org/10.1007/978-981-15-9774-9_84).
- [62] A. Ahmad Khan, S. Alsubai, C. Wechtaisong, A. Almadhor, N. Kryvinska, A. Al Hejaili, and U. Ghulam Mohammad, "CD-FL: Cataract images based disease detection using federated learning," *Comput. Syst. Sci. Eng.*, vol. 47, no. 2, pp. 1733–1750, 2023.
- [63] G. Arslan and C. B. Erdas, "Detection of cataract, diabetic retinopathy and glaucoma eye diseases with deep learning approach," *Intell. Methods Eng. Sci.*, vol. 2, no. 2, pp. 42–47, Jun. 2023.
- [64] S. Faizal, C. A. Rajput, R. Tripathi, B. Verma, M. R. Prusty, and S. S. Korade, "Automated cataract disease detection on anterior segment eye images using adaptive thresholding and fine tuned inception-v3 model," *Biomed. Signal Process. Control*, vol. 82, Apr. 2023, Art. no. 104550.
- [65] N. Varma, S. Yadav, and J. K. P. S. Yadav, "A reliable automatic cataract detection using deep learning," *Int. J. Syst. Assurance Eng. Manage.*, vol. 14, no. 3, pp. 1089–1102, Jun. 2023, doi: [10.1007/s13198-023-01923-2](https://doi.org/10.1007/s13198-023-01923-2).
- [66] J. K. P. S. Yadav and S. Yadav, "Computer-aided diagnosis of cataract severity using retinal fundus images and deep learning," *Comput. Intell.*, vol. 38, no. 4, pp. 1450–1473, Aug. 2022.
- [67] J. Manikandan, B. V. Krishna, C. Sasivarma, R. Y. Kumar, and M. S. Raj, "Cataract fundus image detection using hybrid deep learning model," in *Computational Intelligence in Data Science (Advances in Information and Communication Technology)*, vol. 673, K. R. Chandran and H. Hameed, Eds., Cham, Switzerland: Springer, 2023, doi: [10.1007/978-3-031-38296-3_23](https://doi.org/10.1007/978-3-031-38296-3_23).
- [68] S. Yadav and J. K. P. S. Yadav, "Automatic cataract severity detection and grading using deep learning," *J. Sensors*, vol. 2023, no. 1, Jan. 2023, Art. no. 2973836.
- [69] A. Jones, K. Abisheek, R. D. Kumar, and M. Madesh, "Cataract detection using deep convolutional neural networks," in *Soft Computing and Signal Processing (Advances in Intelligent Systems and Computing)*, vol. 1413, V. S. Reddy, V. K. Prasad, J. Wang, and K. Reddy, Eds., Singapore: Springer, 2022, doi: [10.1007/978-981-16-7088-6_46](https://doi.org/10.1007/978-981-16-7088-6_46).
- [70] T. Pratap and P. Kokil, "Computer-aided diagnosis of cataract using deep transfer learning," *Biomedical Signal Process. Control*, vol. 53, Aug. 2019, Art. no. 101533.
- [71] S. Sahayam, J. Silambarasan, and U. Jayaraman, "Detection of cataract from fundus images using deep transfer learning," in *Computer Vision and Image Processing (Communications in Computer and Information Science)*, vol. 1567, B. Raman, S. Murala, A. Chowdhury, A. Dhali, and A. P. Goyal, Eds., Cham, Switzerland: Springer, 2022, doi: [10.1007/978-3-031-11346-8_16](https://doi.org/10.1007/978-3-031-11346-8_16).
- [72] K. Chauhan, K. Dagar, and R. K. Yadav, "Cataract detection from eye fundus image using an ensemble of transfer learning models," in *Proc. 2nd Int. Conf. Advance Comput. Innov. Technol. Eng. (ICACITE)*, Greater Noida, India, Apr. 2022, pp. 2194–2198, doi: [10.1109/ICACITE53722.2022.9823638](https://doi.org/10.1109/ICACITE53722.2022.9823638).
- [73] R. Angeline, R. Vani, A. J. Sonali, and D. A. Rao, "Automated detection of cataract using a deep learning technique," in *Computational Intelligence in Machine Learning (Lecture Notes in Electrical Engineering)*, vol. 834, A. Kumar, J. M. Zurada, V. K. Gunjan, and R. Balasubramanian, Eds., Singapore: Springer, 2022, doi: [10.1007/978-981-16-8484-5_38](https://doi.org/10.1007/978-981-16-8484-5_38).
- [74] M. Bosch, C. M. Gifford, D. P. Harvie, G. W. Cibis, and A. Agah, "Transfer learning for early detection and classification of amblyopia," *Proc. SPIE*, vol. 11137, Sep. 2019, Art. no. 1113702, doi: [10.1117/12.2523524](https://doi.org/10.1117/12.2523524).
- [75] K. Praveen, S. Lalitha, and S. Gopinath, "A detection of amblyopia using image processing and machine learning techniques," *J. Comput. Sci. Eng. Softw. Test.*, vol. 5, no. 2, pp. 1–11, 2023, doi: [10.5281/zenodo.3066525](https://doi.org/10.5281/zenodo.3066525).
- [76] K. Murali, V. Krishna, V. Krishna, and B. Kumari, "Application of deep learning and image processing analysis of photographs for amblyopia screening," *Indian J. Ophthalmol.*, vol. 68, no. 7, pp. 1407–1410, 2020, doi: [10.4103/ijo.IJO_1399_19](https://doi.org/10.4103/ijo.IJO_1399_19).
- [77] S. Lalitha, N. Shanthi, and S. Gopinath, "A detection of amblyopia medical condition in biomedical datasets using image segmentation and detection processing," *J. Med. Imag. Health Informat.*, vol. 11, no. 11, pp. 2814–2821, Nov. 2021.
- [78] Y. Fan, L. Li, P. Chu, Q. Wu, Y. Wang, W. Cao, and N. Li, "Clinical analysis of eye movement-based data in the medical diagnosis of amblyopia," *Methods*, vol. 213, pp. 26–32, May 2023.
- [79] A. Devlieger, A. Youssfi, and M. Cordonnier, "Evaluation of the bling vision screener in the detection of amblyopia and strabismus in children," *Translational Vis. Sci. Technol.*, vol. 11, no. 4, p. 10, Apr. 2022, doi: [10.1167/tvst.11.4.10](https://doi.org/10.1167/tvst.11.4.10).
- [80] S. B. Aoun, T. Sammouda, W. Walha, M. Sahli, and A. Trigui, "Role of artificial intelligence in predicting visual prognosis after open globe trauma in children," *AlQalam J. Med. Appl. Sci.*, vol. 6, no. 1, pp. 127–133, Mar. 2023. [Online]. Available: <https://www.journal.utripoli.edu.ly/index.php/Alqalam/article/view/187>
- [81] N. Sengar, R. C. Joshi, M. K. Dutta, and R. Burget, "EyeDeep-net: A multi-class diagnosis of retinal diseases using deep neural network," *Neural Comput. Appl.*, vol. 35, no. 14, pp. 10551–10571, May 2023, doi: [10.1007/s00521-023-08249-x](https://doi.org/10.1007/s00521-023-08249-x).
- [82] S. Kadry, V. Rajinikanth, R. González Crespo, and E. Verdú, "Automated detection of age-related macular degeneration using a pre-trained deep-learning scheme," *J. Supercomput.*, vol. 78, no. 5, pp. 7321–7340, Apr. 2022, doi: [10.1007/s11227-021-04181-w](https://doi.org/10.1007/s11227-021-04181-w).
- [83] M. Treder, J. L. Lauermann, and N. Eter, "Automated detection of exudative age-related macular degeneration in spectral domain optical coherence tomography using deep learning," *Graefes Arch. Clin. Experim. Ophthalmol.*, vol. 256, no. 2, pp. 259–265, Feb. 2018, doi: [10.1007/s00417-017-3850-3](https://doi.org/10.1007/s00417-017-3850-3).
- [84] J. Han, S. Choi, J. I. Park, J. S. Hwang, J. M. Han, H. J. Lee, J. Ko, J. Yoon, and D. D.-J. Hwang, "Classifying neovascular age-related macular degeneration with a deep convolutional neural network based on optical coherence tomography images," *Sci. Rep.*, vol. 12, no. 1, p. 2232, Feb. 2022, doi: [10.1038/s41598-022-05903-7](https://doi.org/10.1038/s41598-022-05903-7).
- [85] D. Ma, M. Kumar, V. Khetan, P. Sen, M. Bhende, S. Chen, T. T. L. Yu, S. Lee, E. V. Navajas, J. A. Matsubara, M. J. Ju, M. V. Sarunic, R. Raman, and M. F. Beg, "Clinical explainable differential diagnosis of polypoidal choroidal vasculopathy and age-related macular degeneration using deep learning," *Comput. Biol. Med.*, vol. 143, Apr. 2022, Art. no. 105319.
- [86] T. He, Q. Zhou, and Y. Zou, "Automatic detection of age-related macular degeneration based on deep learning and local outlier factor algorithm," *Diagnostics*, vol. 12, no. 2, p. 532, Feb. 2022, doi: [10.3390/diagnostics12020532](https://doi.org/10.3390/diagnostics12020532).

- [87] N. Motozawa, G. An, S. Takagi, S. Kitahata, M. Mandai, Y. Hiram, H. Yokota, M. Akiba, A. Tsujikawa, M. Takahashi, and Y. Kurimoto, "Optical coherence tomography-based deep-learning models for classifying normal and age-related macular degeneration and exudative and non-exudative age-related macular degeneration changes," *Ophthalmol. Therapy*, vol. 8, no. 4, pp. 527–539, Dec. 2019, doi: [10.1007/s40123-019-00207-y](https://doi.org/10.1007/s40123-019-00207-y).
- [88] Y. Peng, S. Dharssi, Q. Chen, T. D. Keenan, E. Agrón, W. T. Wong, E. Y. Chew, and Z. Lu, "DeepSeeNet: A deep learning model for automated classification of patient-based age-related macular degeneration severity from color fundus photographs," *Ophthalmology*, vol. 126, no. 4, pp. 565–575, Apr. 2019.
- [89] D. K. Prasad, L. Vibha, and K. R. Venugopal, "Machine learning based early detection of age-related macular degeneration: Early warning system," in *Proc. 14th Int. Conf. Inf. Process. (ICINPRO)*, Bangalore, India, Dec. 2018, pp. 1–6, doi: [10.1109/ICINPRO43533.2018.9096760](https://doi.org/10.1109/ICINPRO43533.2018.9096760).
- [90] R. Chakraborty and A. Pramanik, "Deep learning-based automated detection of age-related macular degeneration from retinal fundus images," in *Proc. Int. Conf. Frontiers Comput. Syst.*, in *Advances in Intelligent Systems and Computing*, vol. 1255, D. Bhattacharjee, D. K. Kole, N. Dey, S. Basu, D. Plewczynski, Eds., Singapore: Springer, 2021, pp. 433–441, doi: [10.1007/978-981-15-7834-2_41](https://doi.org/10.1007/978-981-15-7834-2_41).
- [91] P. Udayaraju and P. Jeyanthi, "Early diagnosis of age-related macular degeneration (ARMD) using deep learning," in *Intelligent Systems and Sustainable Computing. Smart Innovation, Systems and Technologies*, vol. 289, V. S. Reddy, V. K. Prasad, D. N. M. Rao, and S. C. Satapathy, Eds., Singapore: Springer, 2022, doi: [10.1007/978-981-19-0011-2_59](https://doi.org/10.1007/978-981-19-0011-2_59).
- [92] R. Chakraborty and A. Pramanik, "DCNN-based prediction model for detection of age-related macular degeneration from color fundus images," *Med. Biol. Eng. Comput.*, vol. 60, no. 5, pp. 1431–1448, May 2022, doi: [10.1007/s11517-022-02542-y](https://doi.org/10.1007/s11517-022-02542-y).
- [93] S. Mukherjee, T. De Silva, P. Grisso, H. Wiley, D. L. K. Tieman, A. T. Thavikulwat, E. Chew, and C. Cukras, "Retinal layer segmentation in optical coherence tomography (OCT) using a 3D deep-convolutional regression network for patients with age-related macular degeneration," *Biomed. Opt. Exp.*, vol. 13, no. 6, p. 3195, Jun. 2022.
- [94] A. Choudhary, S. Ahlawat, S. Urooj, N. Pathak, A. Lay-Ekuakille, and N. Sharma, "A deep learning-based framework for retinal disease classification," *Healthcare*, vol. 11, no. 2, p. 212, Jan. 2023, doi: [10.3390/healthcare11020212](https://doi.org/10.3390/healthcare11020212).
- [95] P. Selvakumar and R. ArunPrakash, "An efficient investigation on age-related macular degeneration using deep learning with cloud-based teleophthalmology architecture," *J. Biomaterials Tissue Eng.*, vol. 13, no. 4, pp. 499–512, Apr. 2023.
- [96] M. A. Ali, M. S. Hossain, M. K. Hossain, S. S. Sikder, S. A. Khushbu, and M. Islam, "AMDNet23: Hybrid CNN-LSTM deep learning approach with enhanced preprocessing for age-related macular degeneration (AMD) detection," *Intell. Syst. Appl.*, vol. 21, Mar. 2024, Art. no. 200334.
- [97] S. R. Deepti and P. C. Karthik, "Survey on age-related macular degeneration detection in OCT image," *Int. J. Comput. Appl.*, vol. 46, no. 2, pp. 101–112, Feb. 2024, doi: [10.1080/1206212x.2023.2286032](https://doi.org/10.1080/1206212x.2023.2286032).
- [98] S. S. Mishra, B. Mandal, and N. B. Puhon, "MacularNet: Towards fully automated attention-based deep CNN for macular disease classification," *Social Netw. Comput. Sci.*, vol. 3, no. 2, p. 142, Mar. 2022, doi: [10.1007/s42979-022-01024-0](https://doi.org/10.1007/s42979-022-01024-0).



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